

YFG FCL LEAK TEST.

Operator / Maintenance Manual

Manufactured by:
Paveway Automation Pvt. Ltd.
Sanghvi Steel Complex,
128/2, TELCO Road,
Chinchwad,
Pune - 411019

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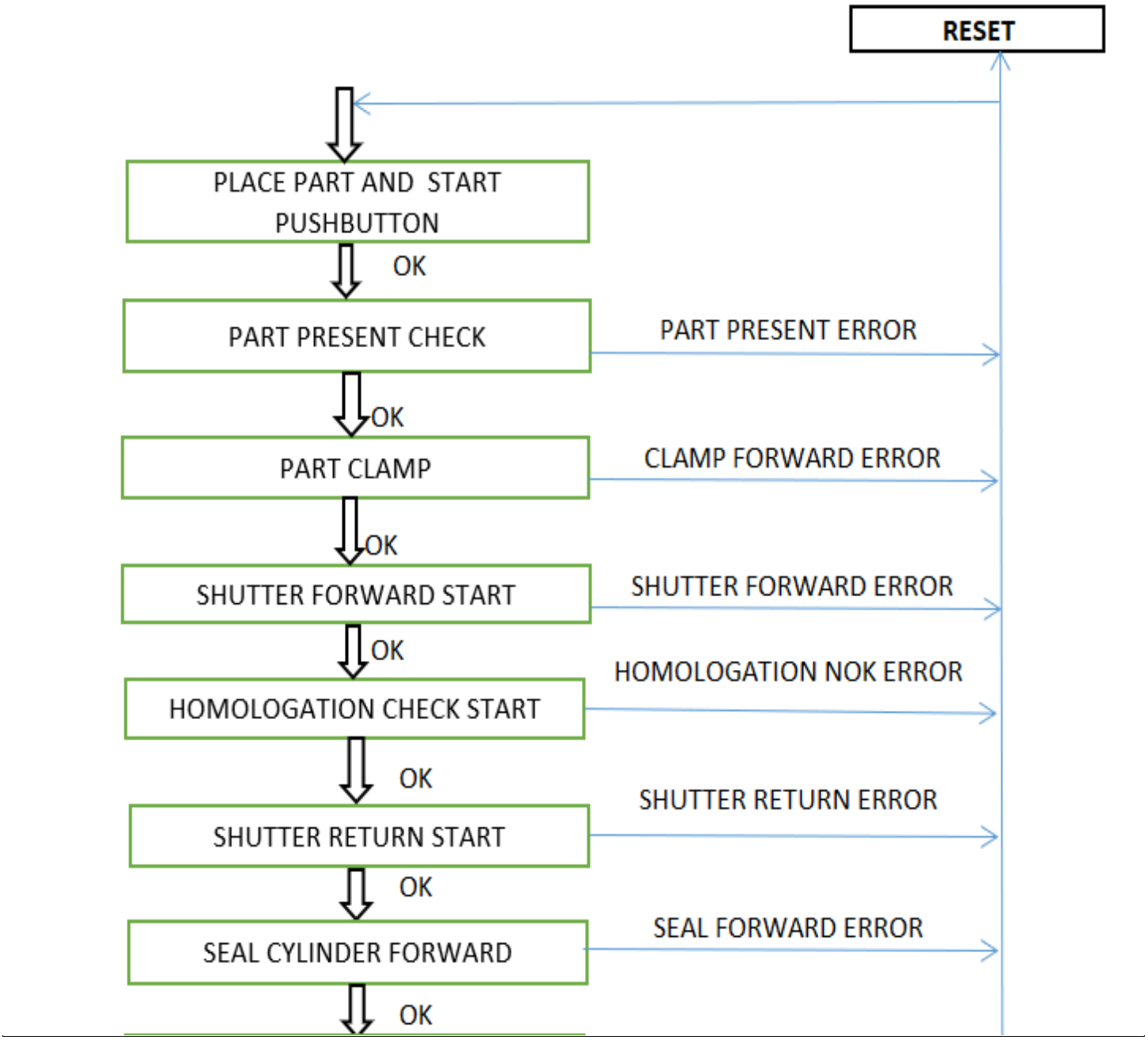
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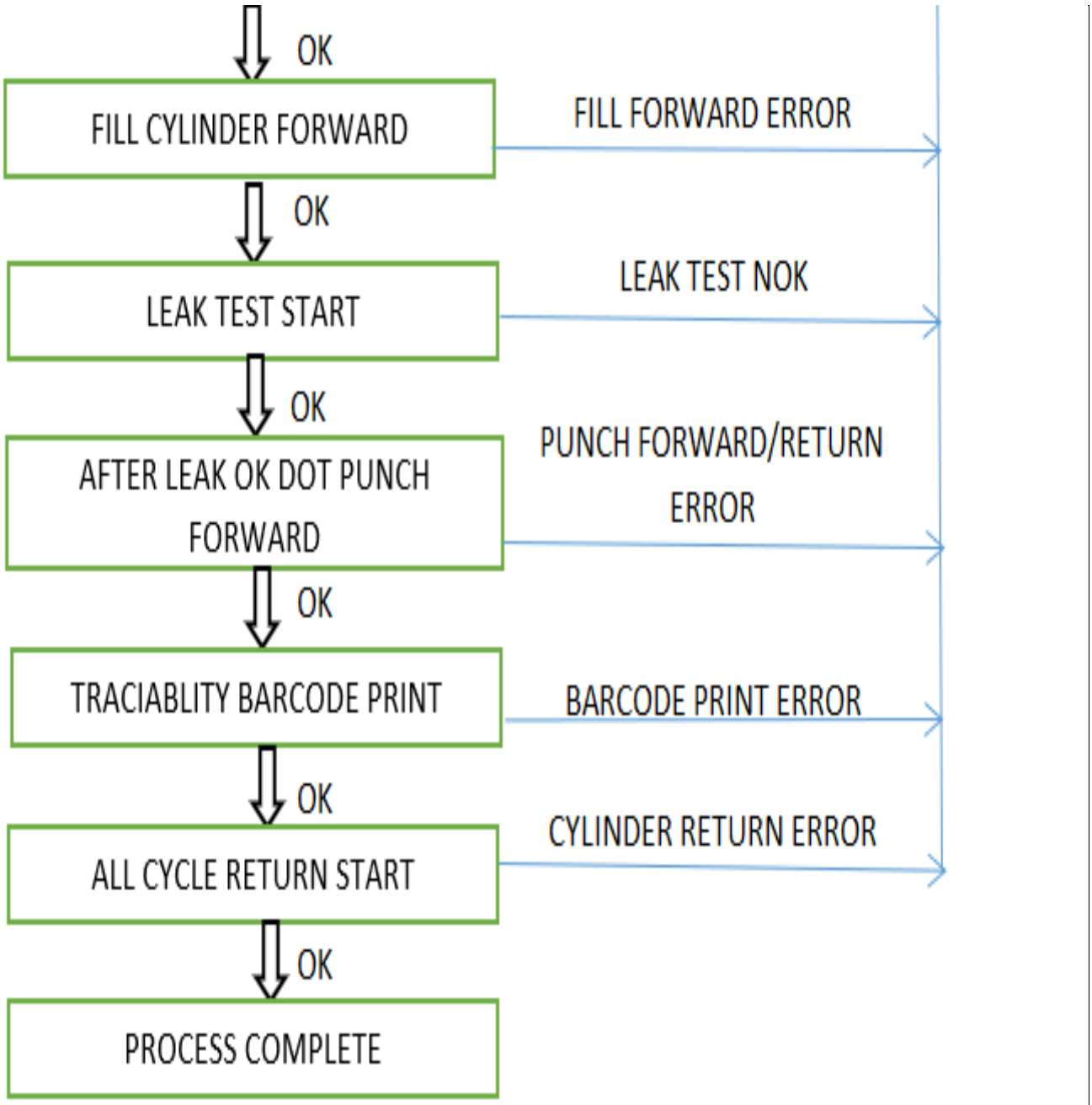
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SYSTEM INFORMATION:

Name :	YFG FCL LEAK TEST.
Customer :	Marelli Motherson Automotive Lighting India Pvt Ltd. Gat No-165/1,165/2 &165/3 Village Ambethan, Dwarka School, Taluka Khed, Chakan, Pune-410501
Manufacturing Date :	April, 2022.
Supply Voltage :	230 VAC, 50/60 Hz
Power consumption:	0.535KW
Manufacturer :	**Paveway Automation Pvt. Ltd.** Sanghvi Steel Complex, 128/2, Telco Road, Chinchwad, Pune- 411019. tel: 020-27450840
Email ID :	***applications@pavewayautomation.com***

PROCESS SEQUENCE:




OK
FILL CYLINDER FORWARD
FILL FORWARD ERROR
OK
LEAK TEST START
LEAK TEST NOK
OK
AFTER LEAK OK DOT PUNCH FORWARD
PUNCH FORWARD/RETURN ERROR
OK
TRACIABLITY BARCODE PRINT
BARCODE PRINT ERROR
OK
ALL CYCLE RETURN START
CYLINDER RETURN ERROR
OK
PROCESS COMPLETE

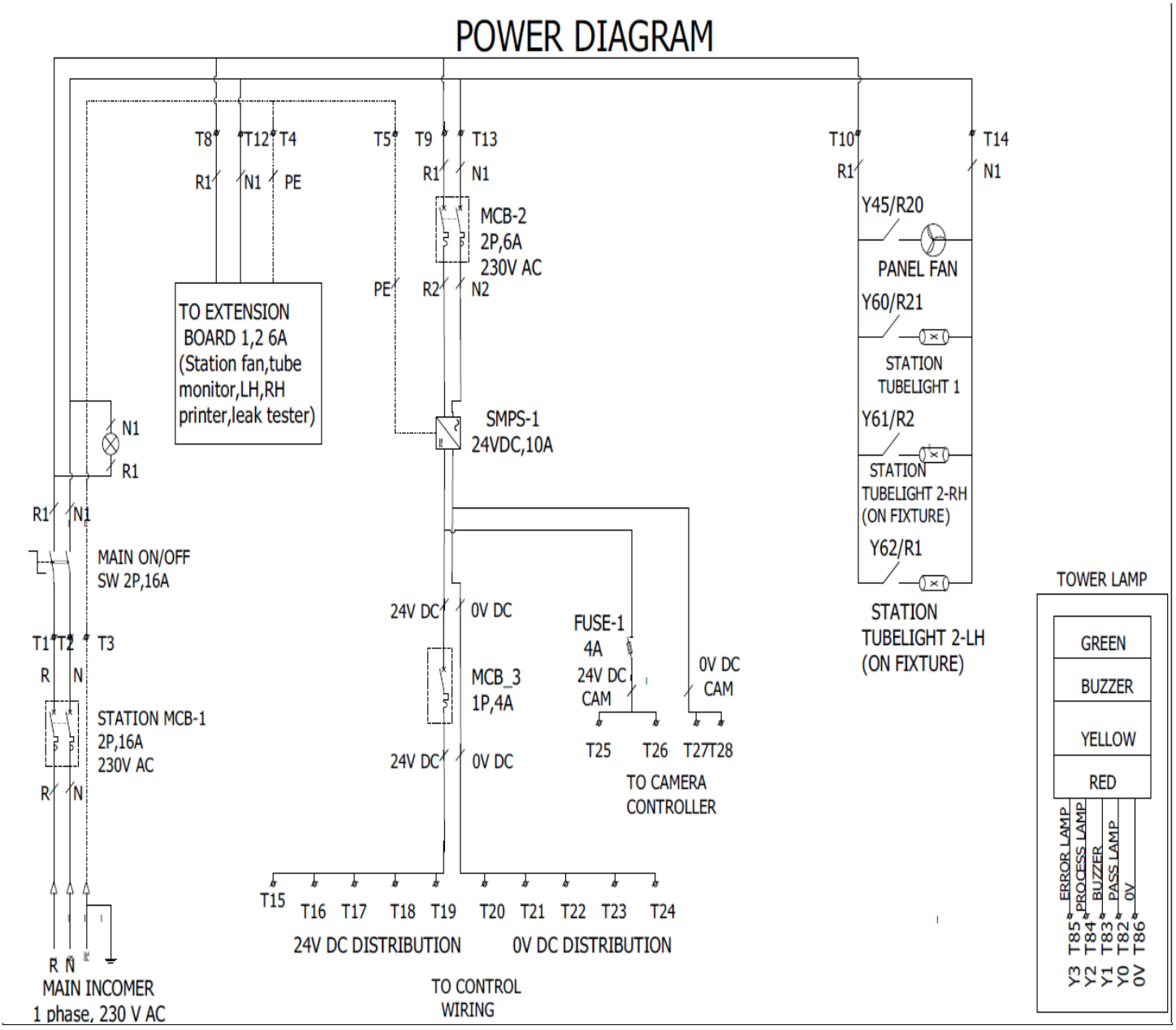
WIRING DIAGRAMS:

SYMBOL DESCRIPTION

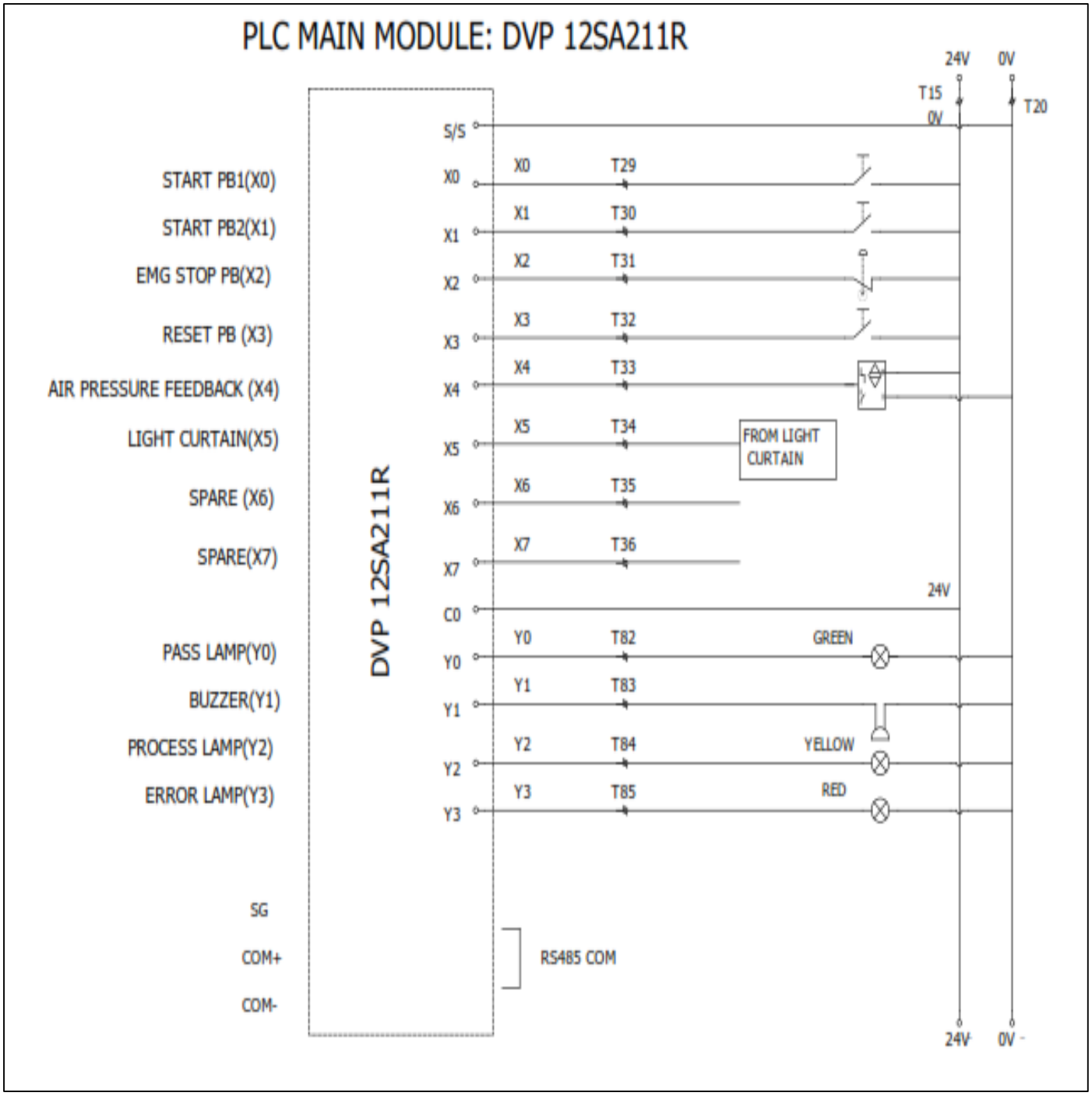
SYMBOL DESCRIPTION

Symbol	Symbol description	Symbol	Symbol description
	Earthing		Proximity sensor
	Isolation Transformer		Airoff
	2 Pole MCB		Buzzer
	2 Pole on/off switch		Double acting solenoid valve
	Lamp		
	Tubelight		
	Fan		
	SMPS		
	1 pole MCB		Double acting solenoid valve (pneumatic)
	Push button		
	Relay coil		
	Emergency stop button		

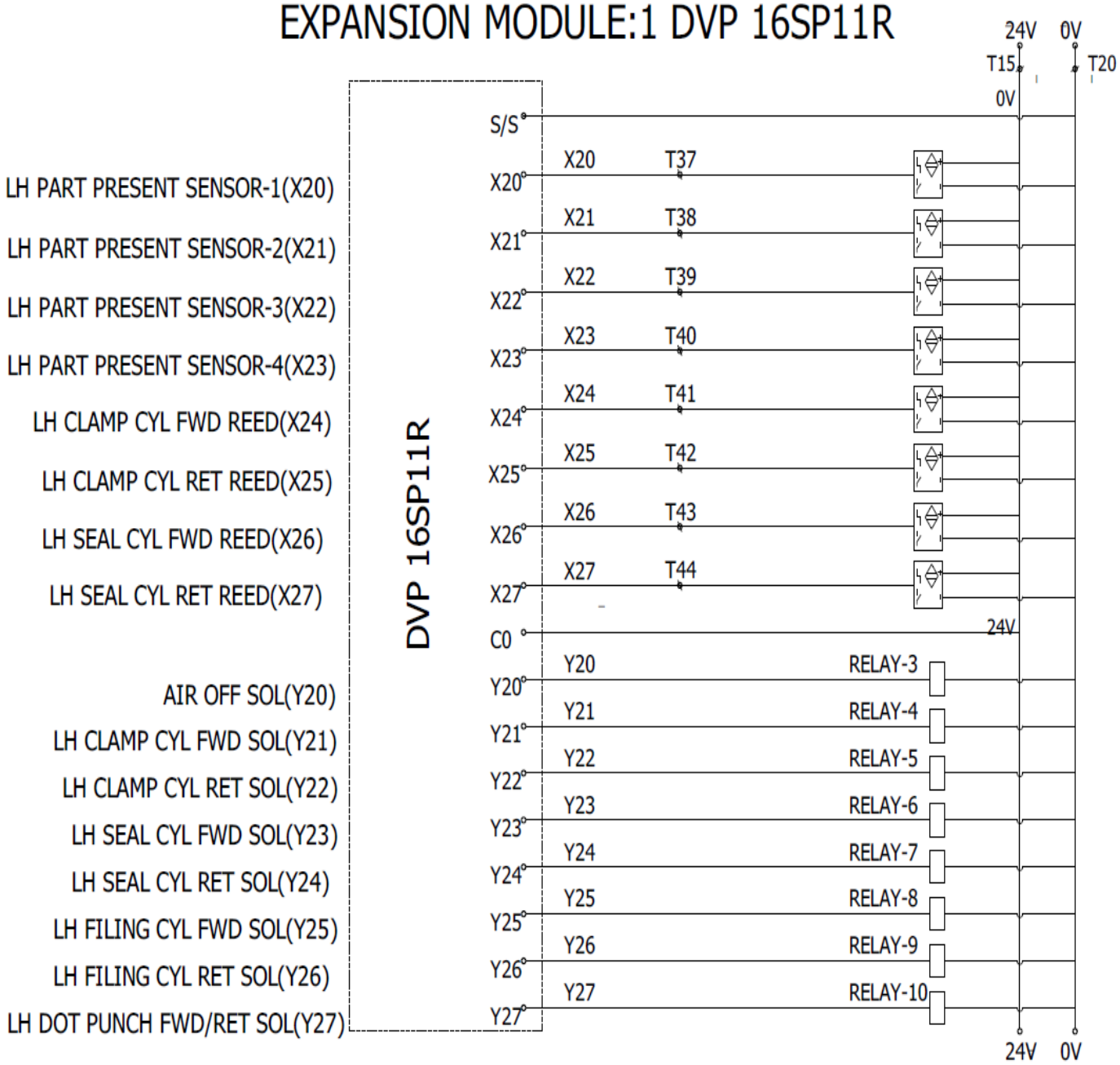
POWER DIAGRAM



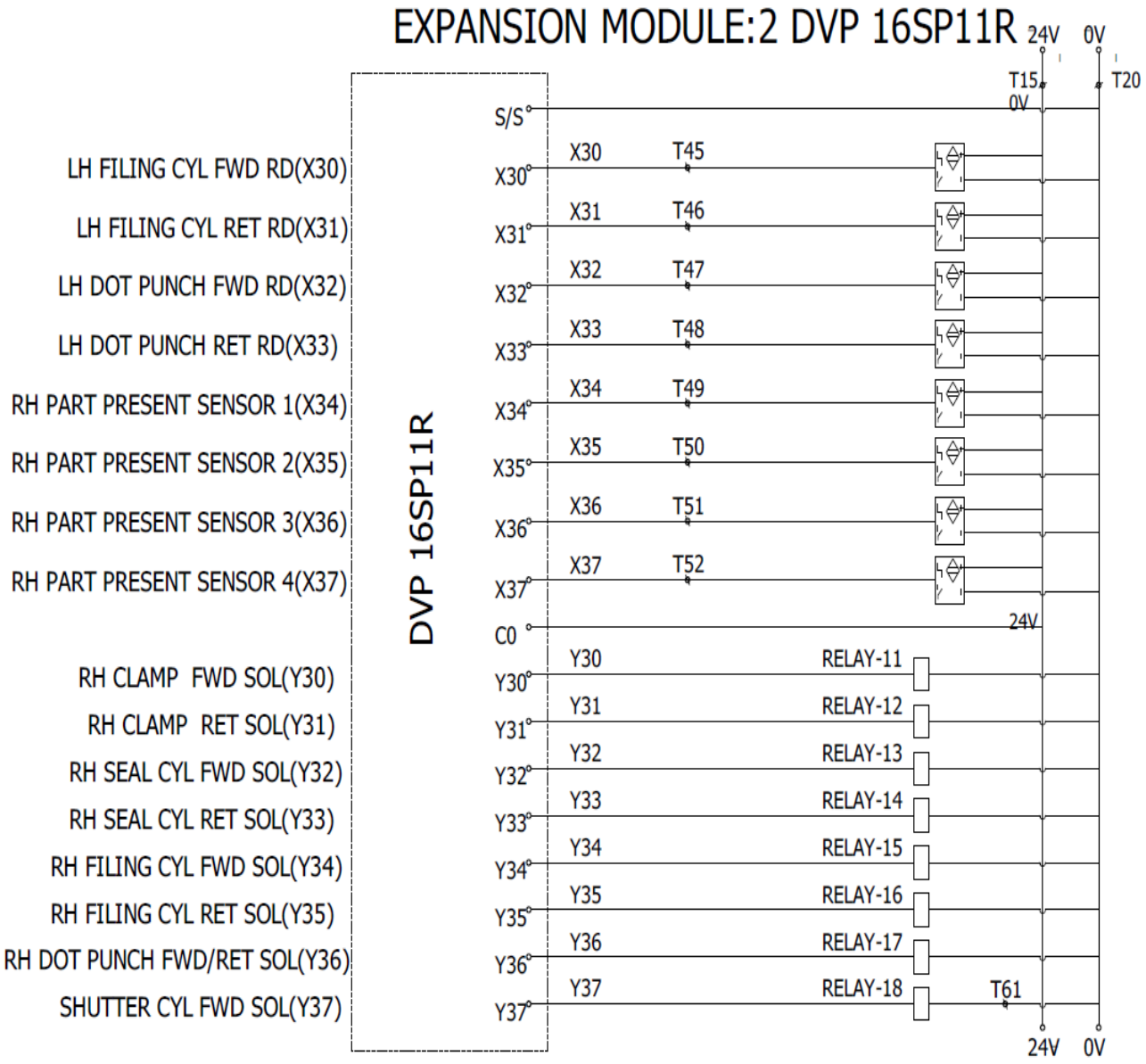
MAIN PLC MODULE-DVP12SA211R



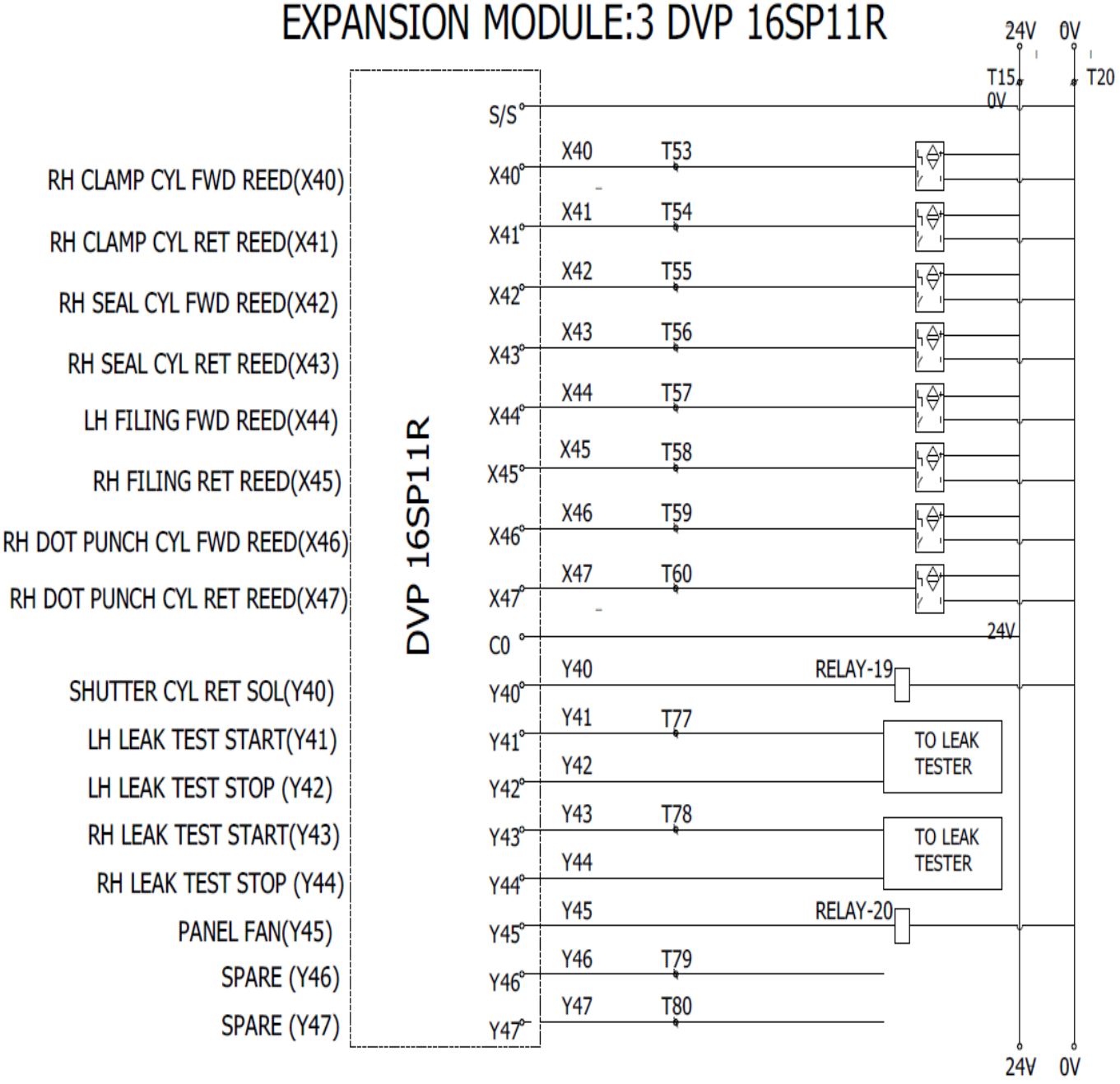
EXPANSION MODULE 1: DVP 16SP11R



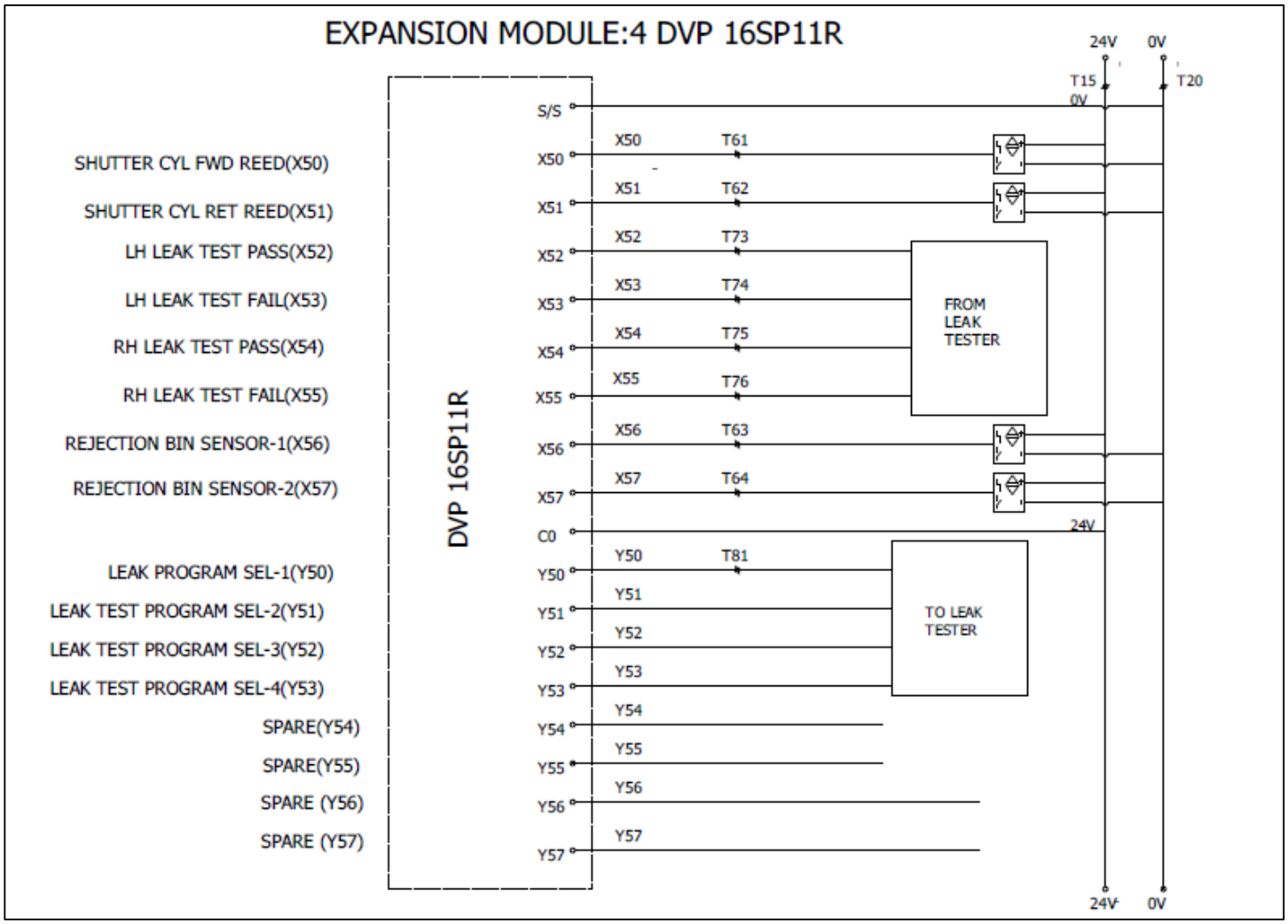
EXPANSION MODULE 2: DVP 16SP11R



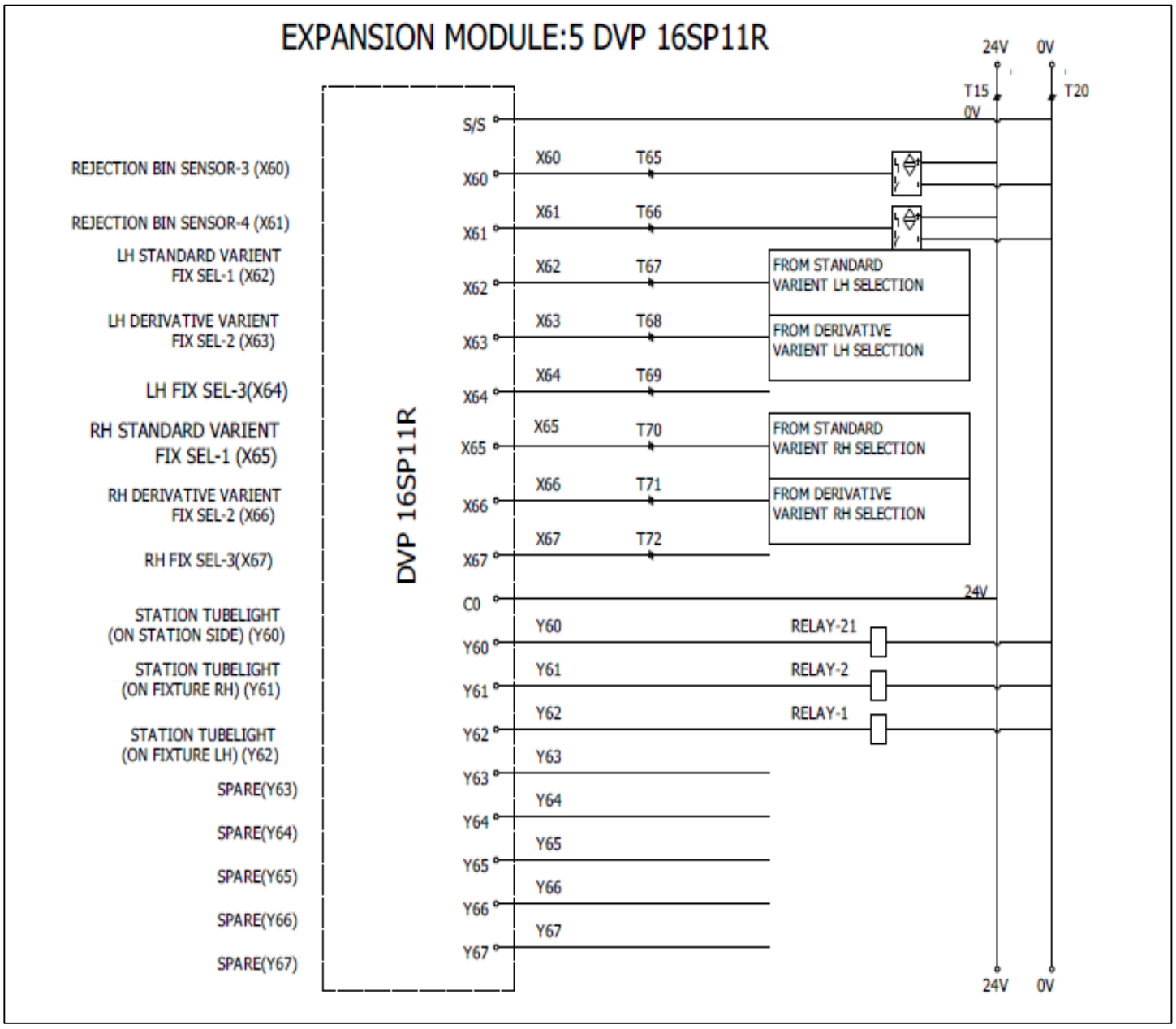
EXPANSION MODULE 3: DVP 16SP11R



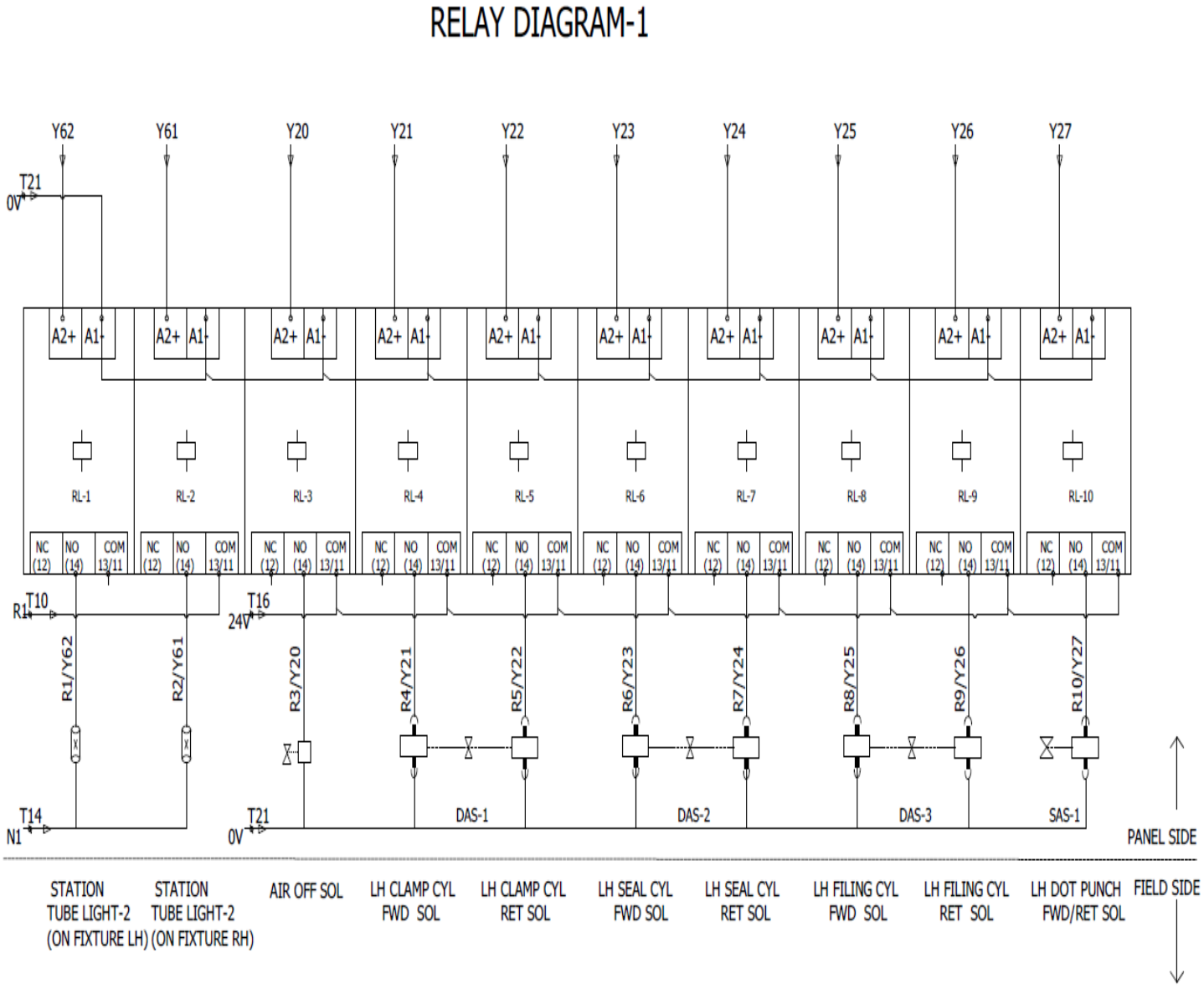
EXPANSION MODULE 4: DVP 16SP11R



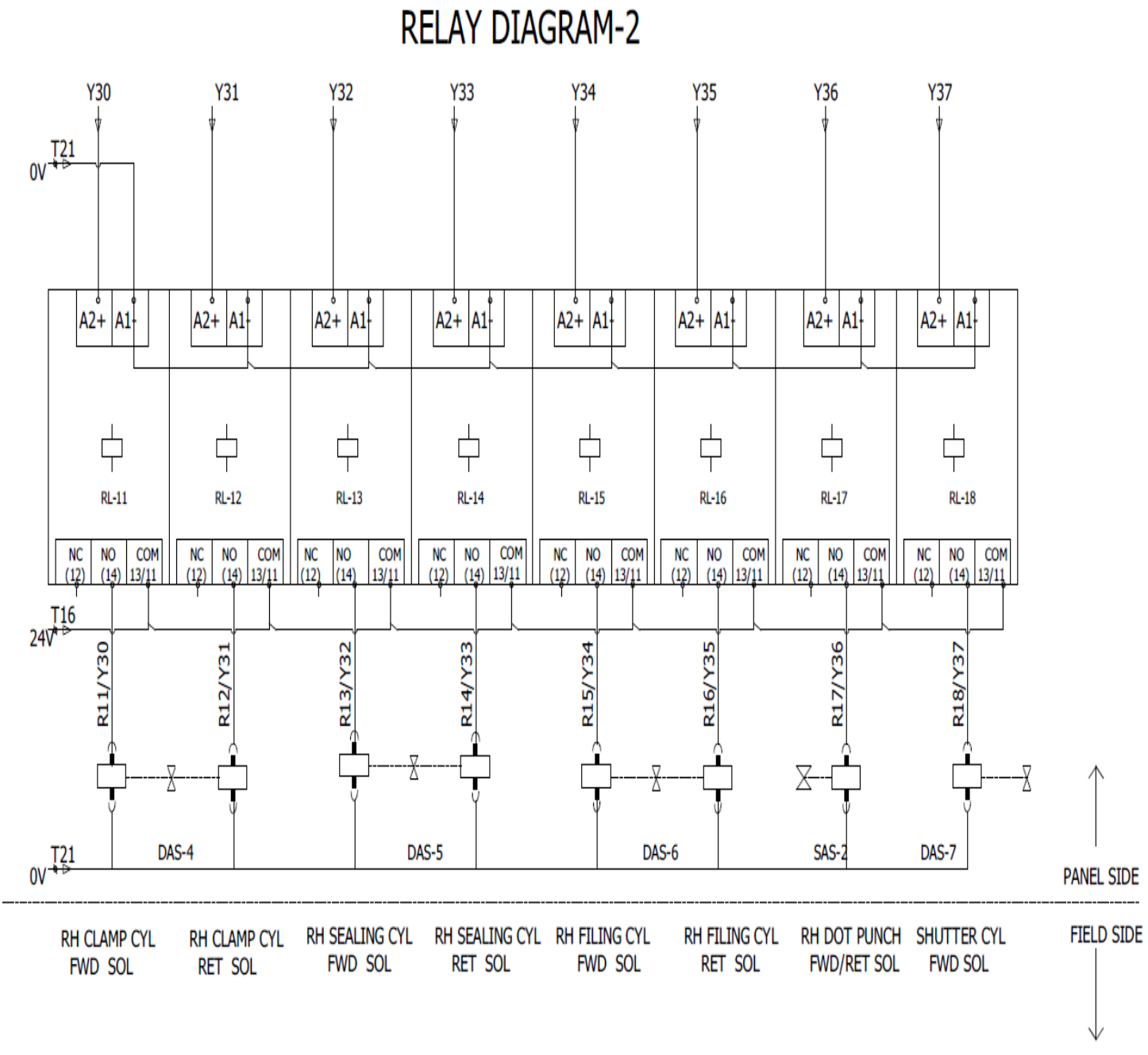
EXPANSION MODULE 5: DVP 16SP11R



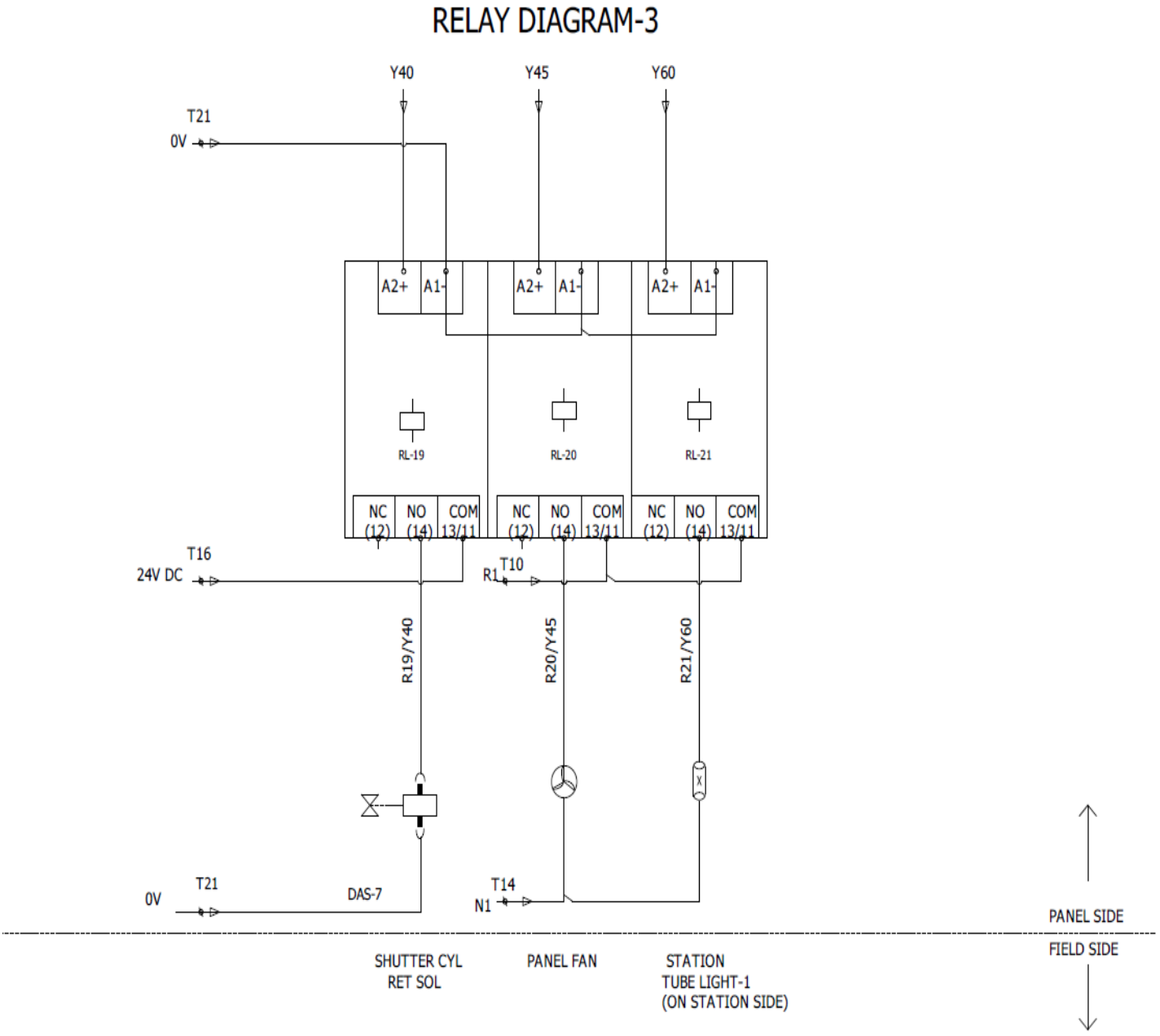
RELAY DIAGRAM-1



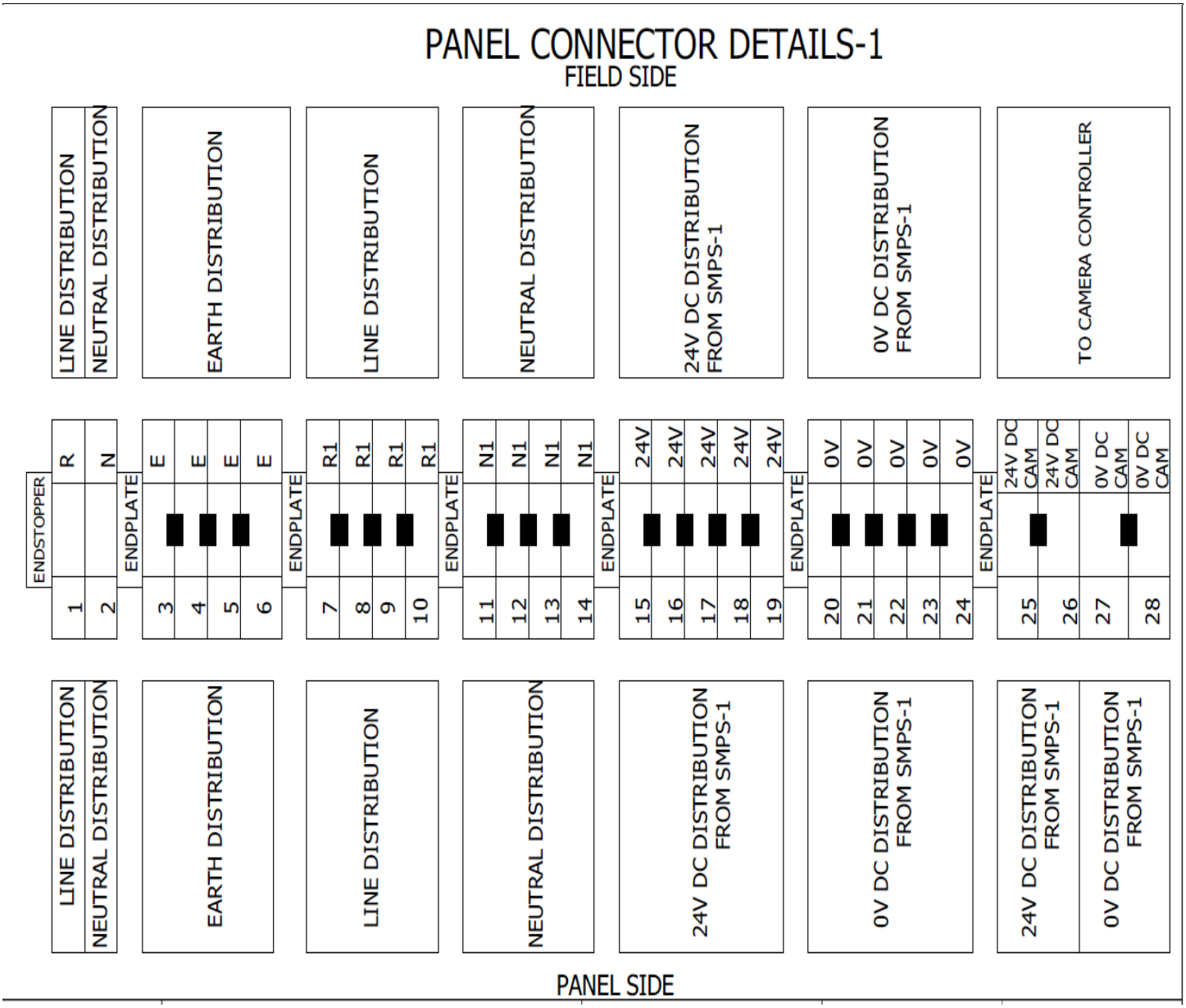
RELAY DIAGRAM-2



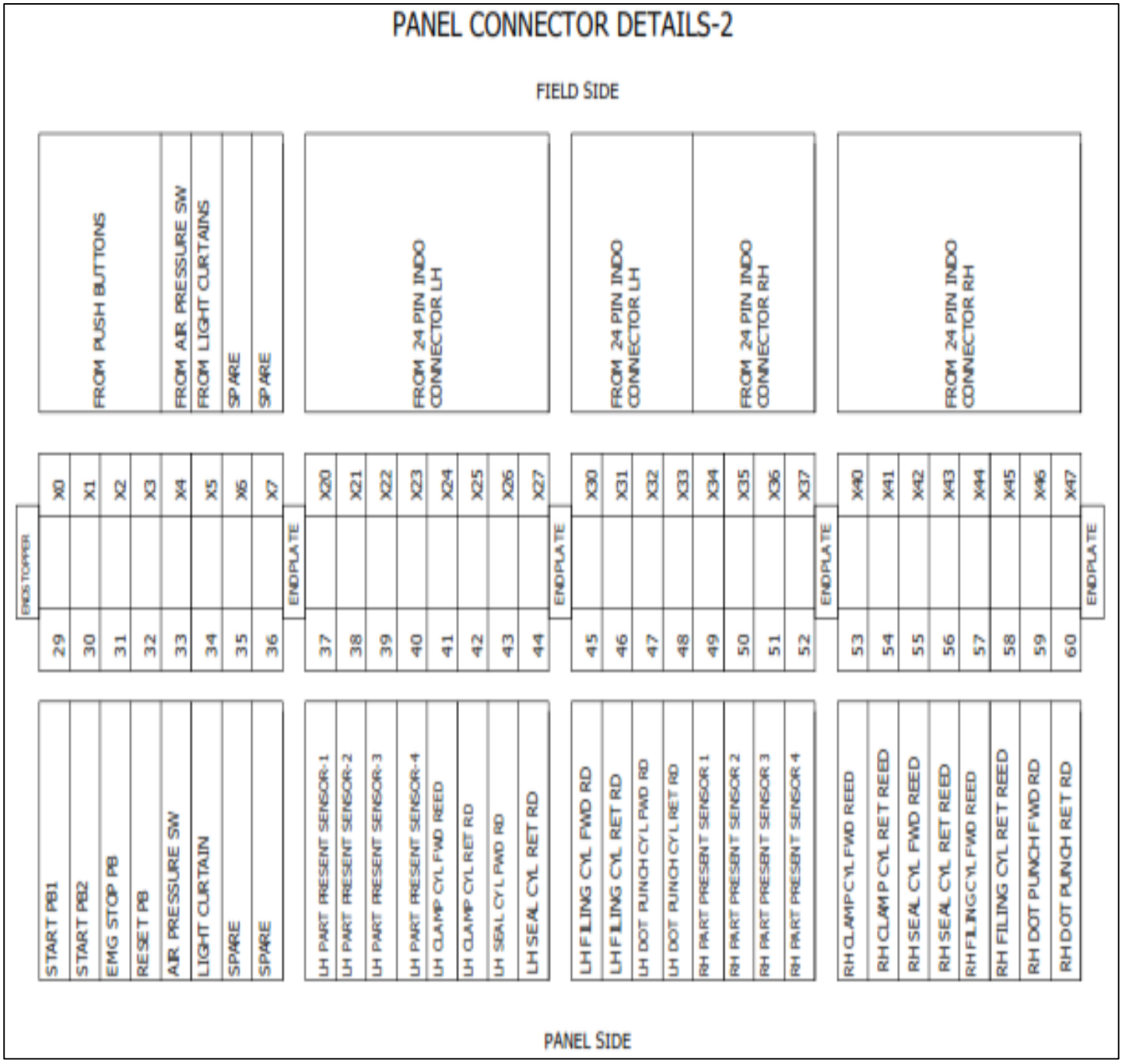
RELAY DIAGRAM-3



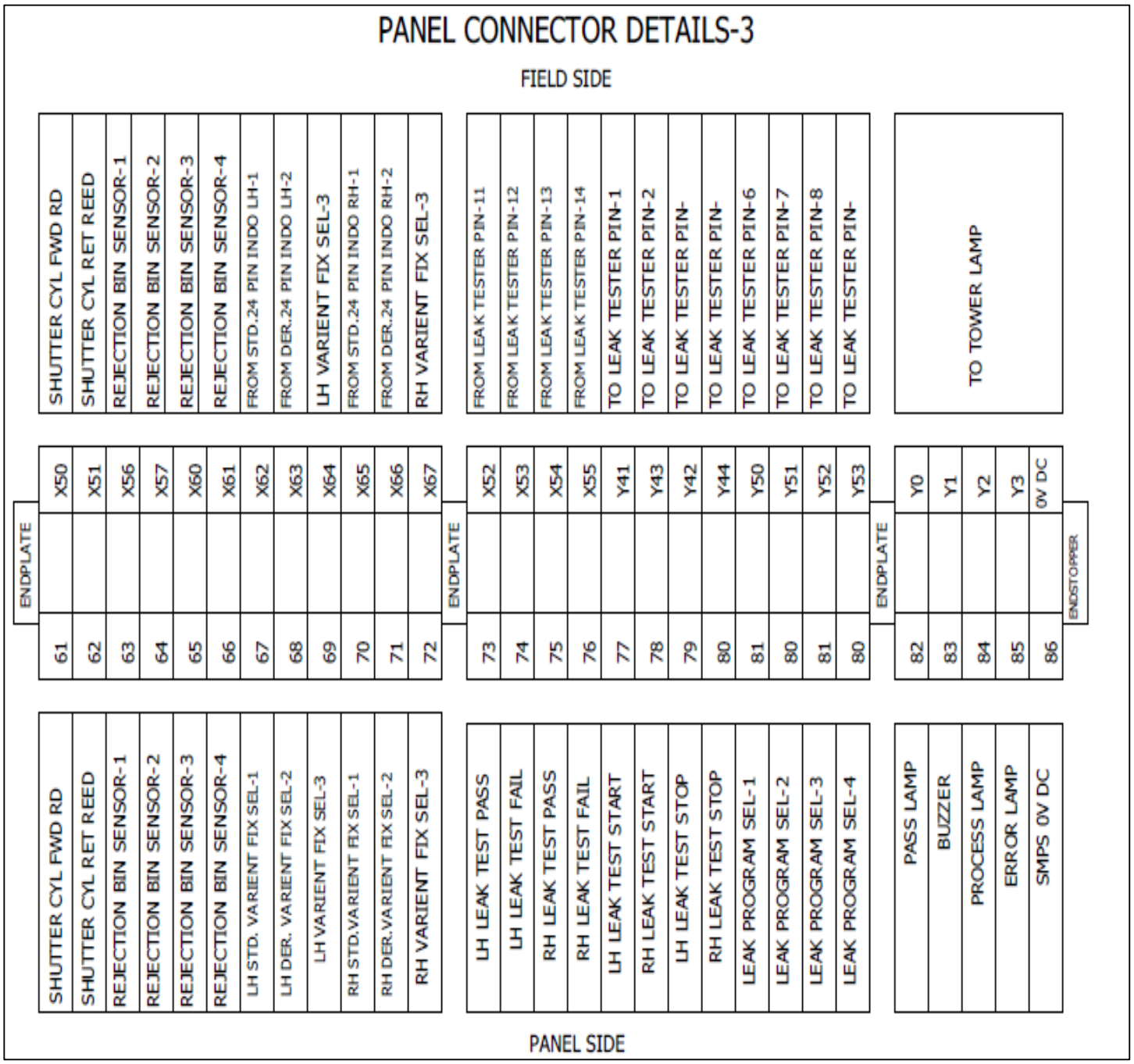
PANEL CONNECTOR DETAILS-1



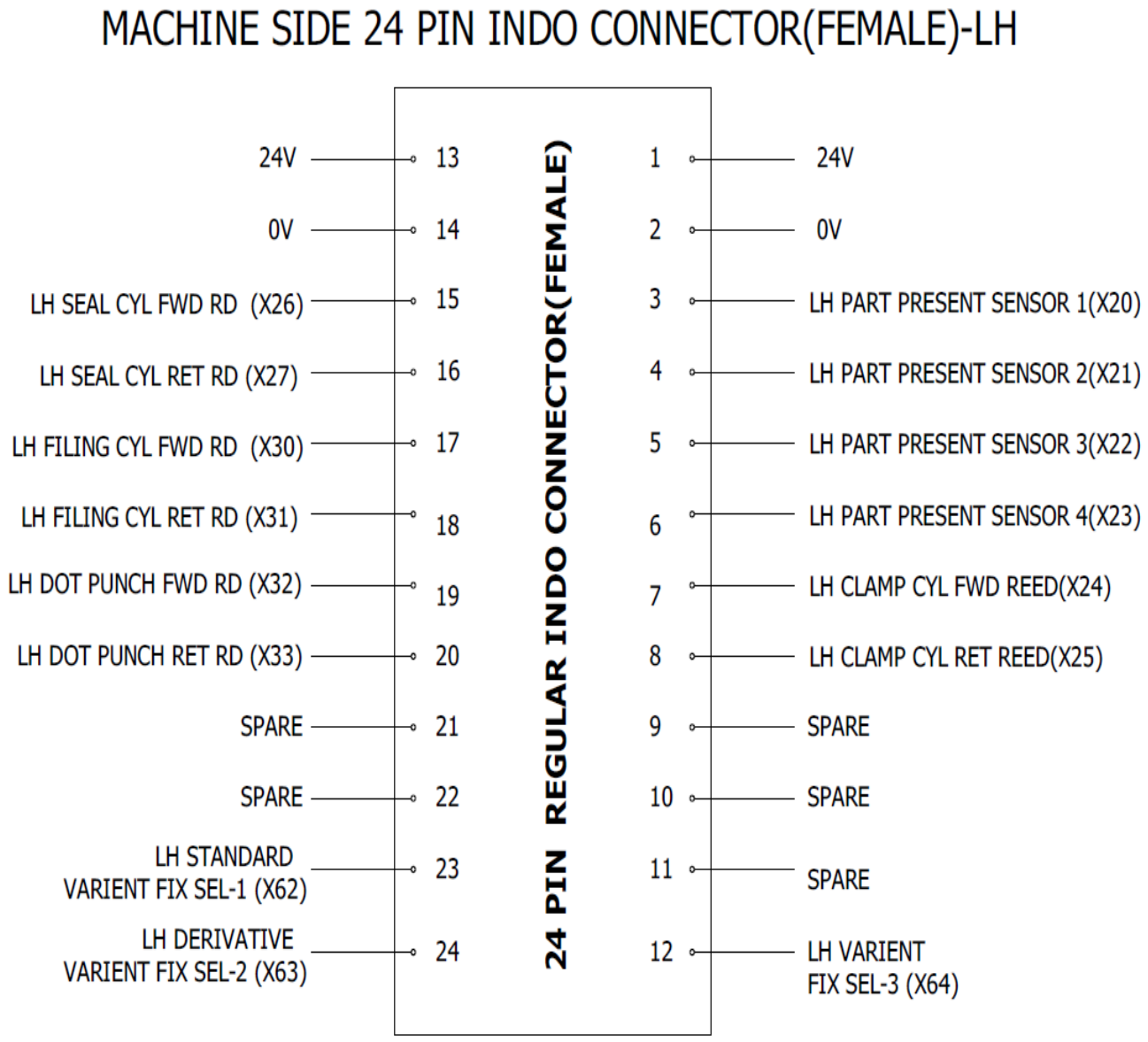
PANEL CONNECTOR DETAILS-2



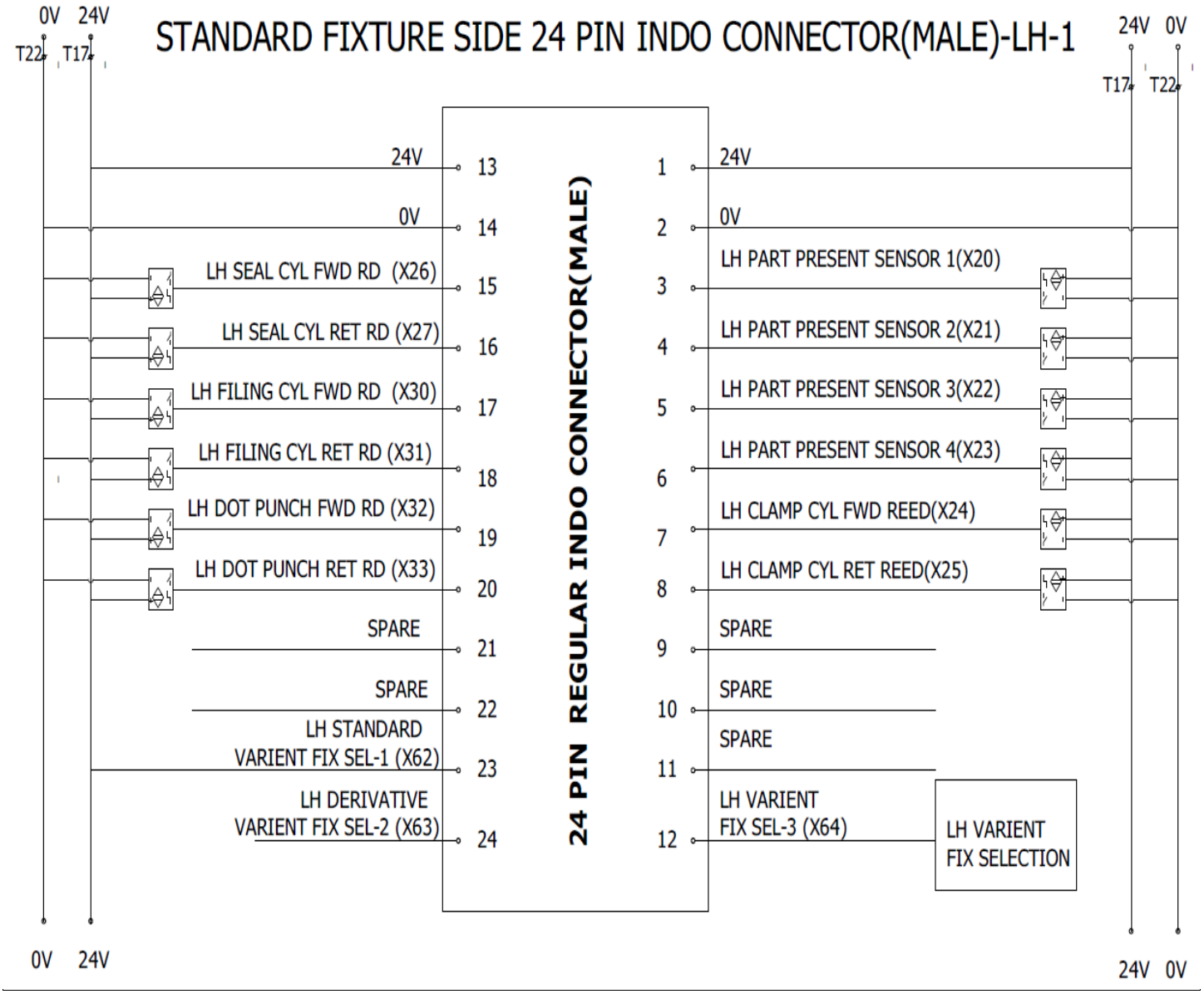
PANEL CONNECTOR DETAILS-3



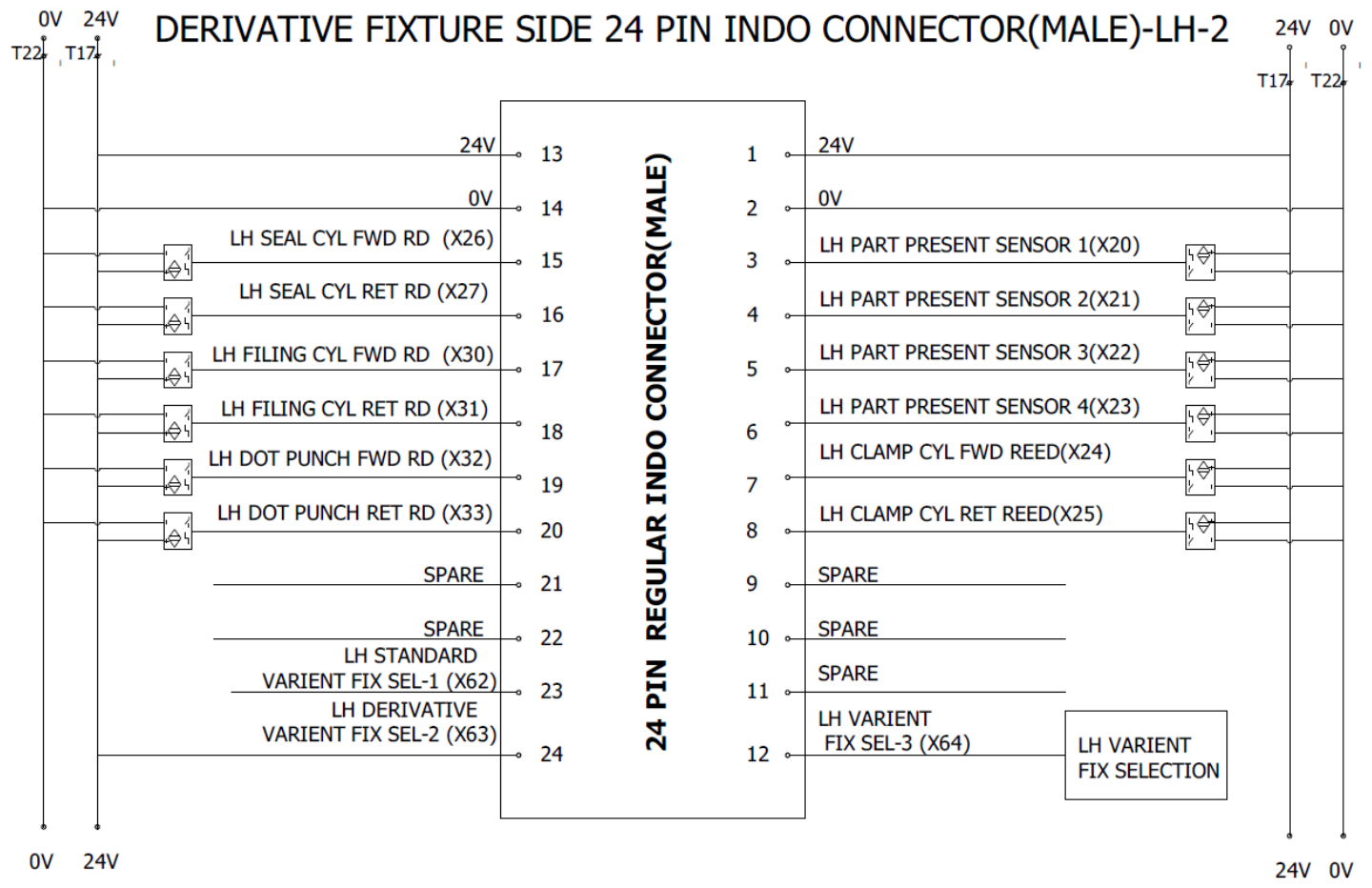
MACHINE SIDE 24 PIN INDO CONNECTOR (FEMALE)



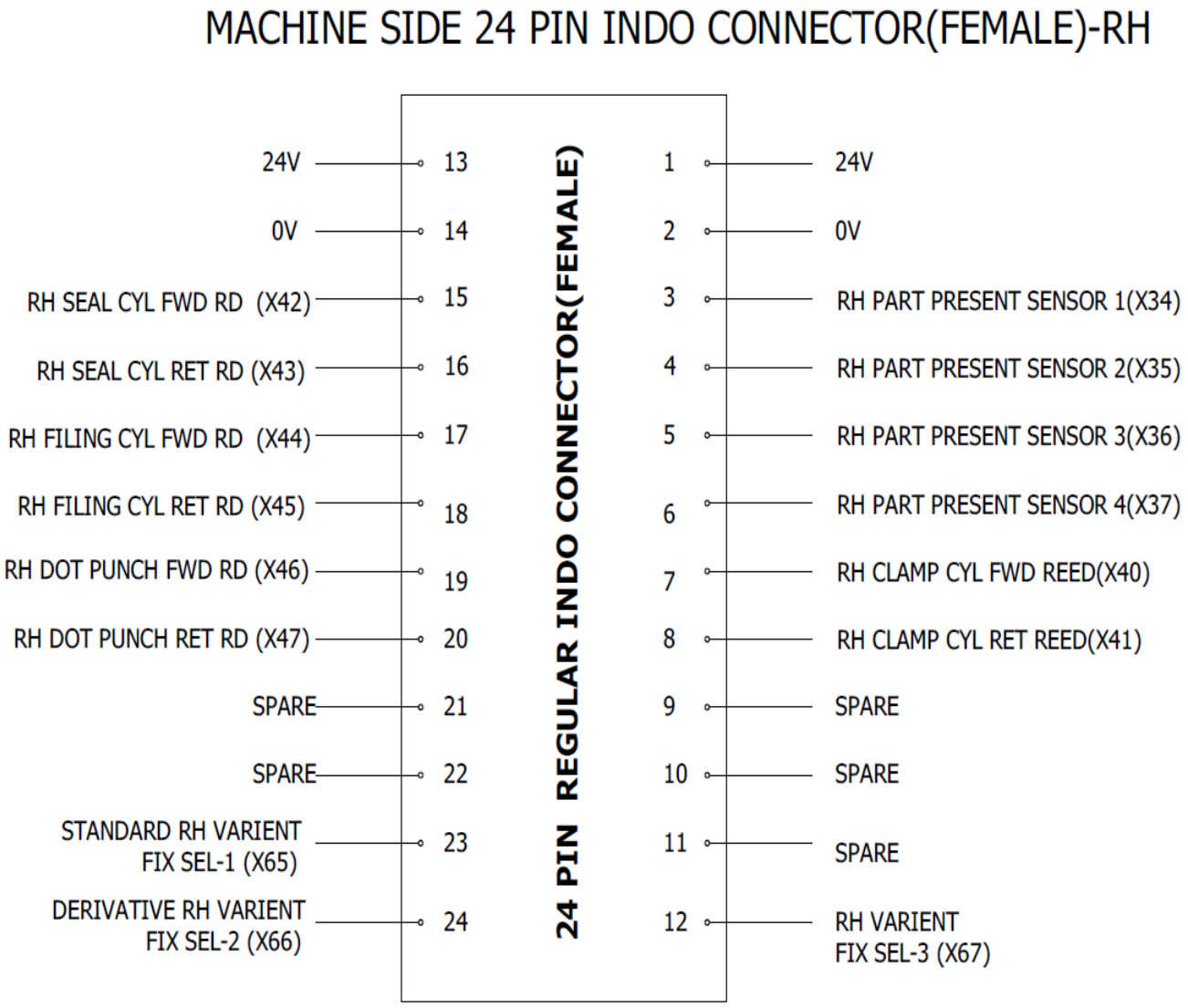
STANDARD FIXTURE SIDE 24 PIN INDO CONNECTOR (MALE) LH-1



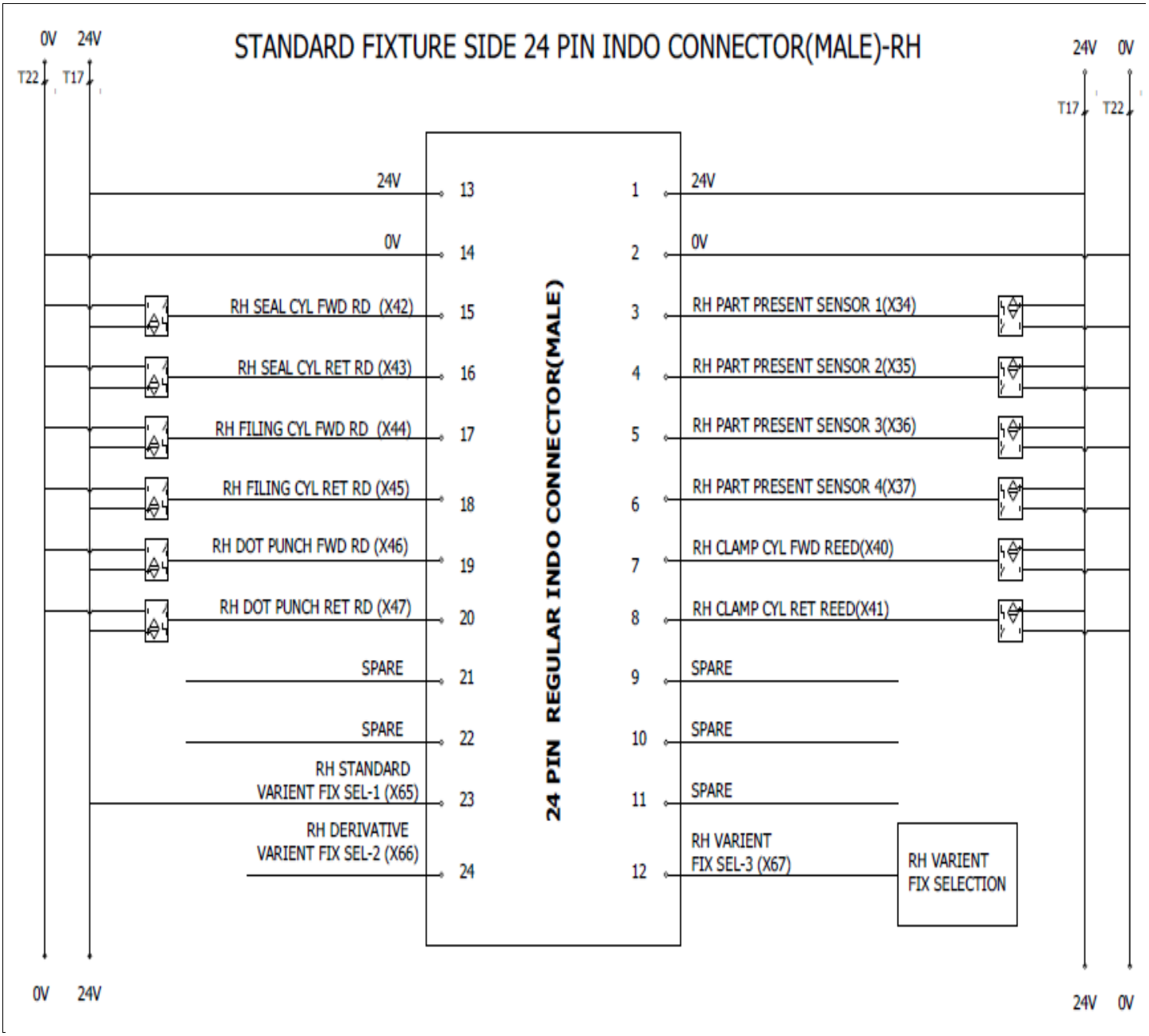
DERIVATIVE FIXTURE SIDE 24 PIN INDO CONNECTOR (MALE) LH-2



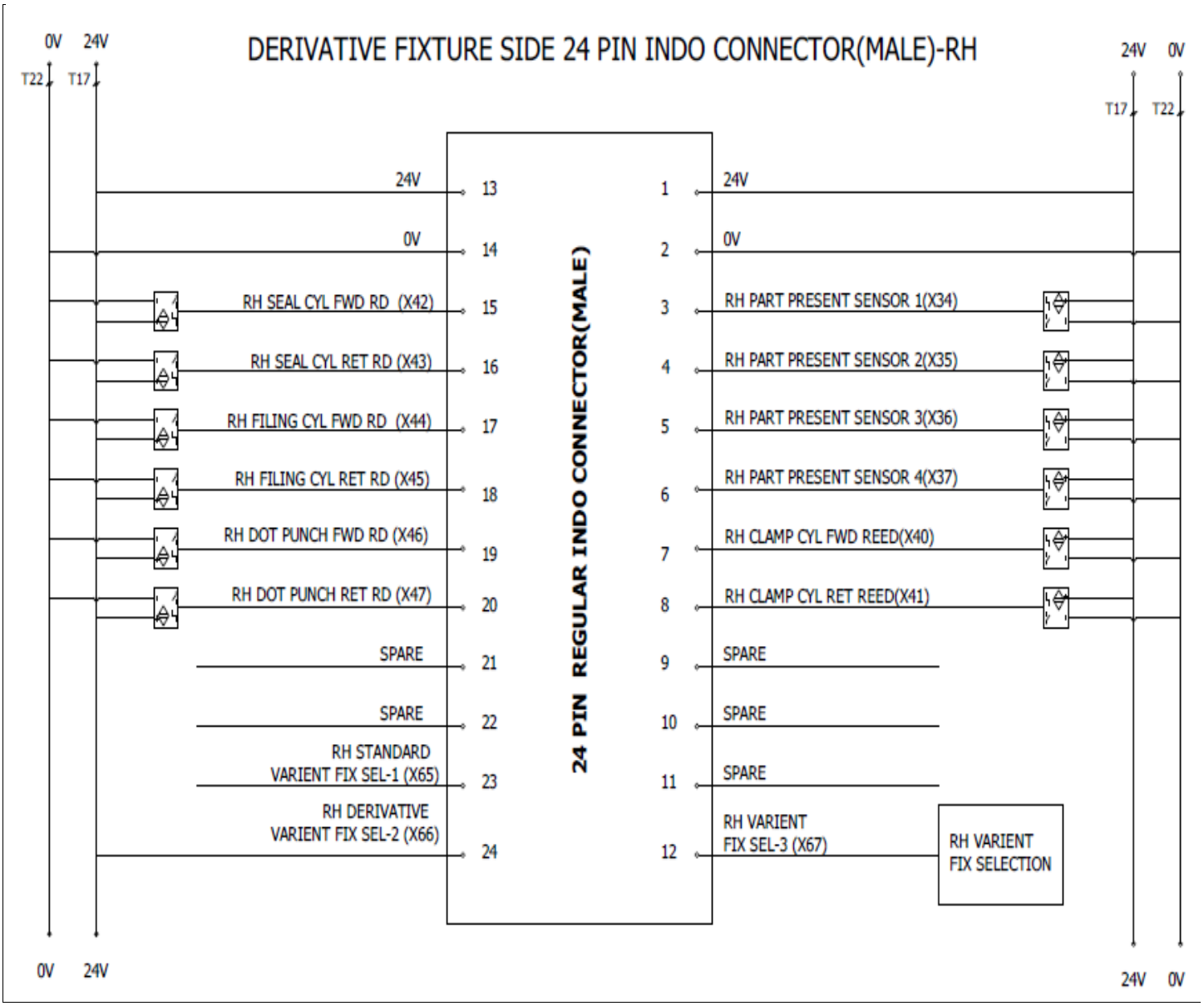
MACHINE SIDE 24 PIN INDO CONNECTOR (FEMALE)-RH



STANDARD FIXTURE SIDE 24 PIN INDO CONNECTOR (MALE) RH-1



DERIVATIVE FIXTURE SIDE 24 PIN INDO CONNECTOR (MALE) RH-2

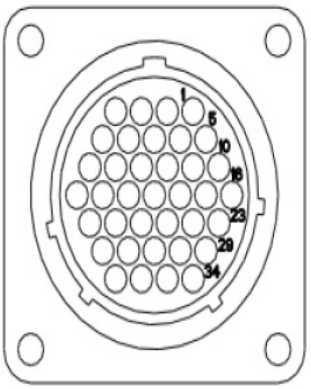


LEAK TESTER 37 I/O PIN CONNECTOR DESCRIPTION (MS2009/2R SERIES)

LEAK TESTER 37 I/O PIN CONNECTOR DESCRIPTION (MS2009/2R SERIES)

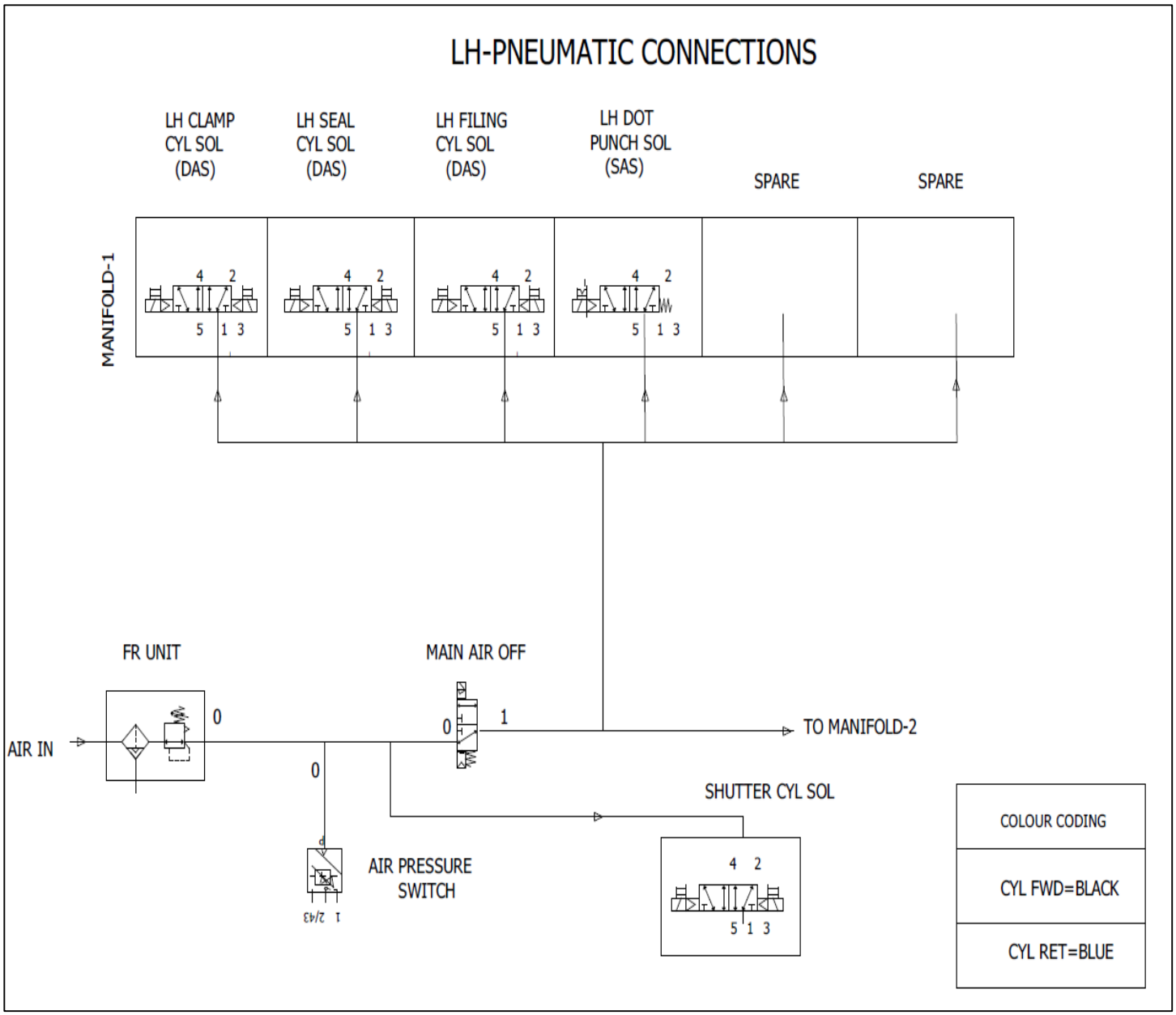
PIN NO.	FUNCTION	DESCRIPTION	PLC I/O ADDRESS
1	IN	START 1	LH LEAK TEST START(Y41)
2	IN	START 2	RH LEAK TEST START(Y43)
3	IN	START 3	-
4	IN	START 4	-
5	IN	LANGUAGE	-
6	IN	PROGRAM SELE BIT A	LEAK PROG SEL-1(Y50)
7	IN	PROGRAM SELE BIT B	LEAK PROG SEL-2(Y51)
8	IN	PROGRAM SELE BIT C	LEAK PROG SEL-3(Y52)
9	PWR	0 V – To be connected with PLC 0 V	0V SMPS
10	PWR	0 V – To be connected with PLC 0 V	0V SMPS
11	OUT	GOOD 1	LH LEAK TEST PASS(X52)
12	OUT	REJECTED 1	LH LEAK TEST FAIL(X53)
13	OUT	GOOD 2	RH LEAK TEST PASS(X54)
14	OUT	REJECTED 2	RH LEAK TEST FAIL(X55)

Instrument connector

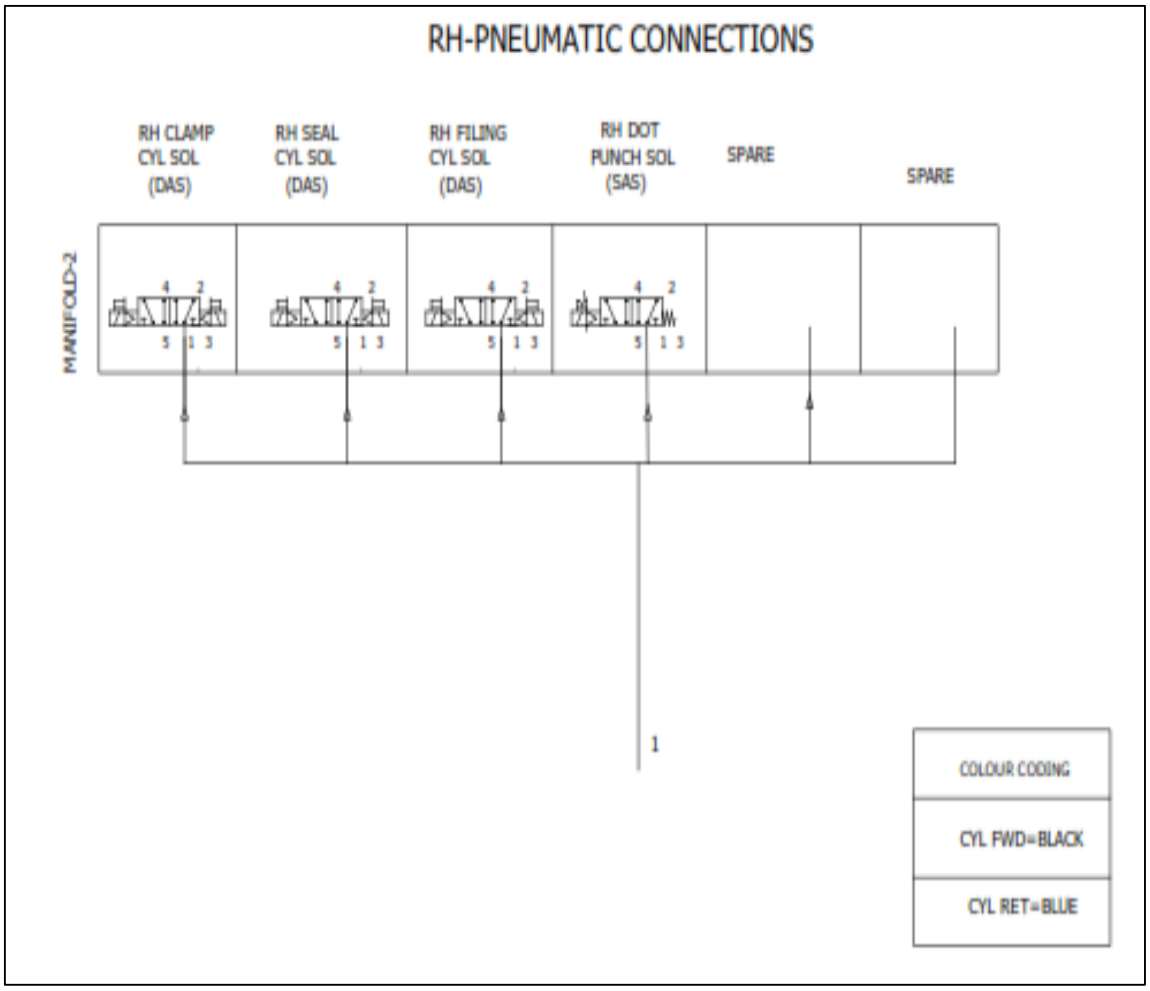


Note:Pin 15 to 37 are spare.

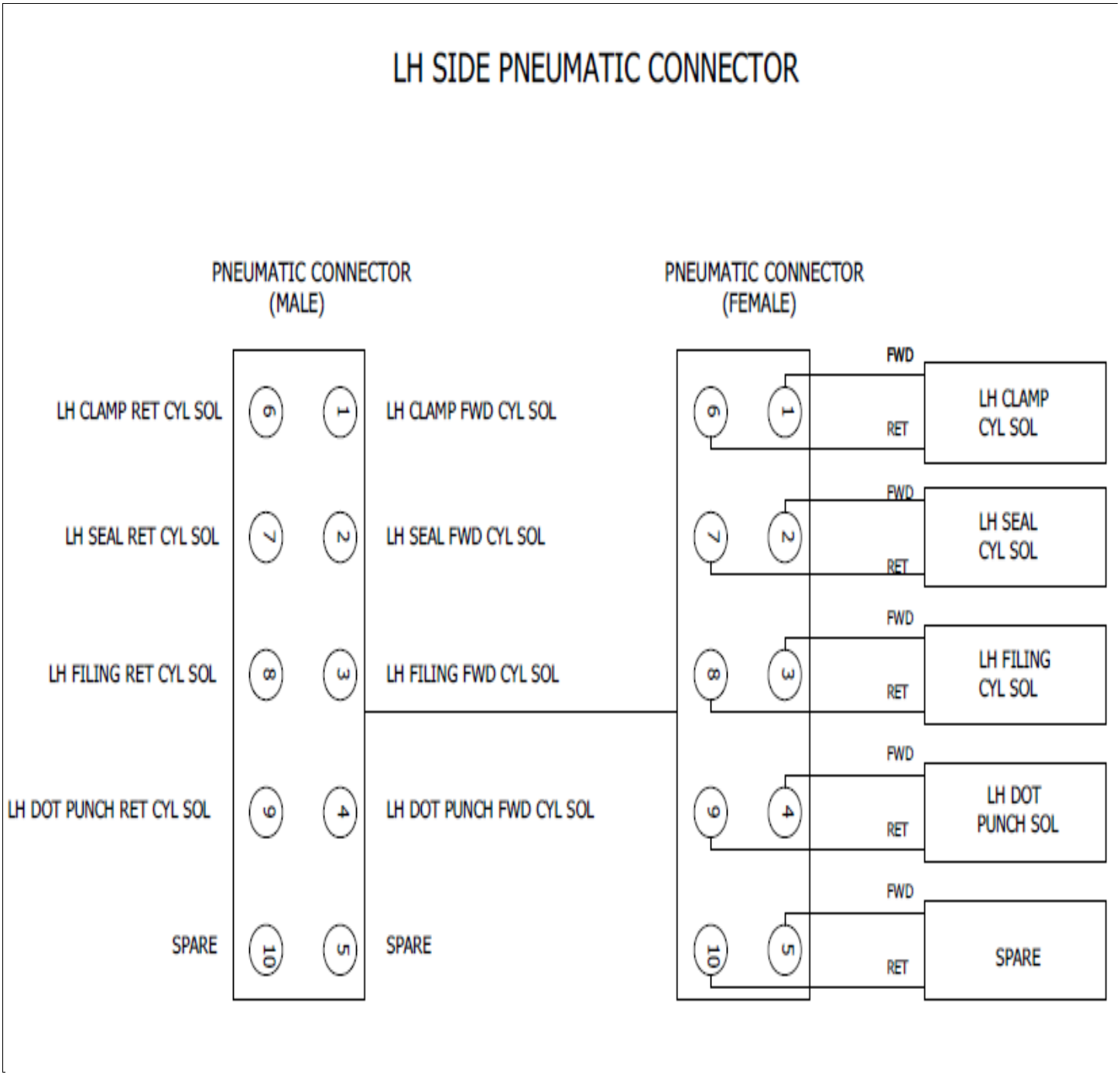
LH-PNEUMATIC CONNECTIONS



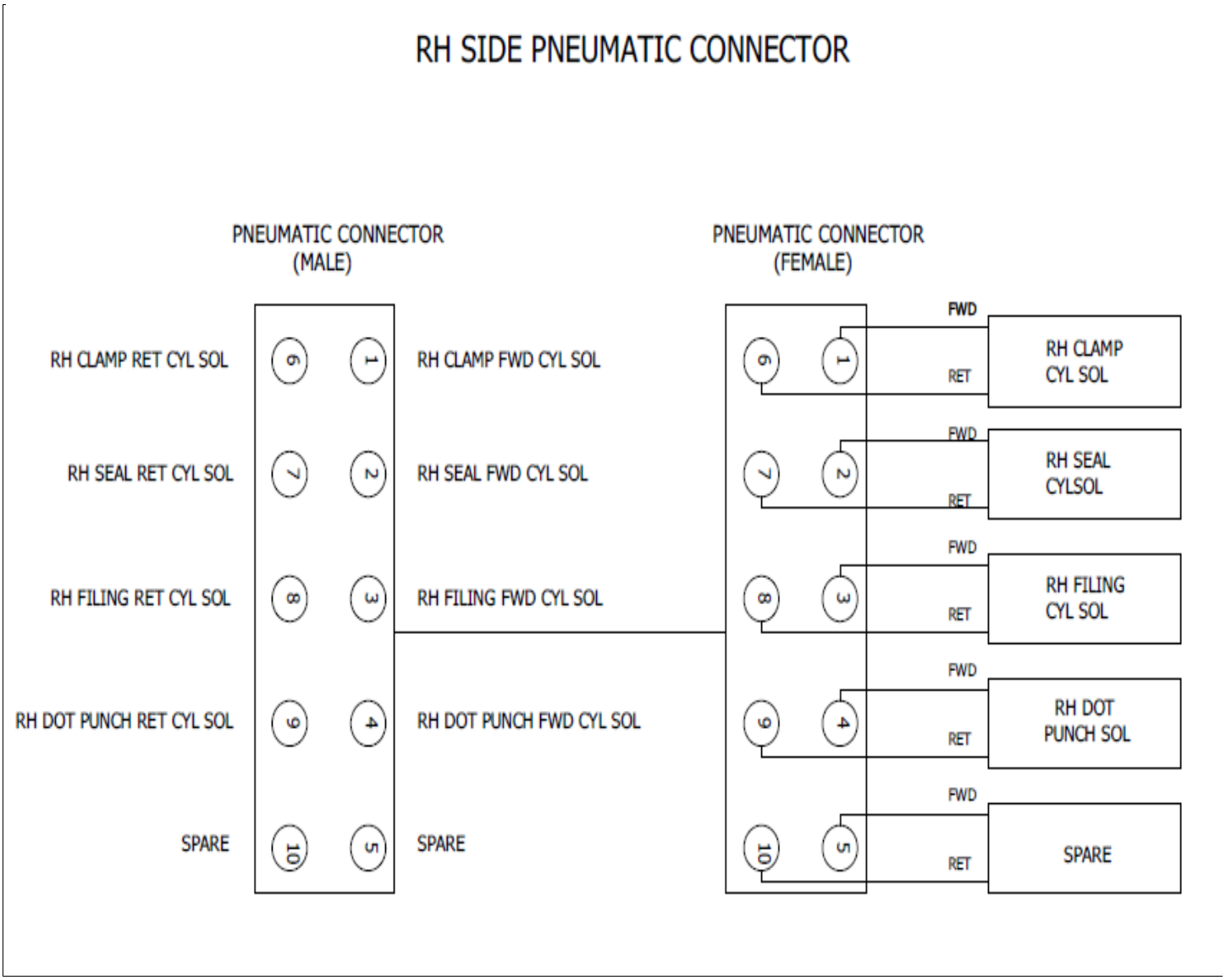
RH-PNEUMATIC CONNECTIONS



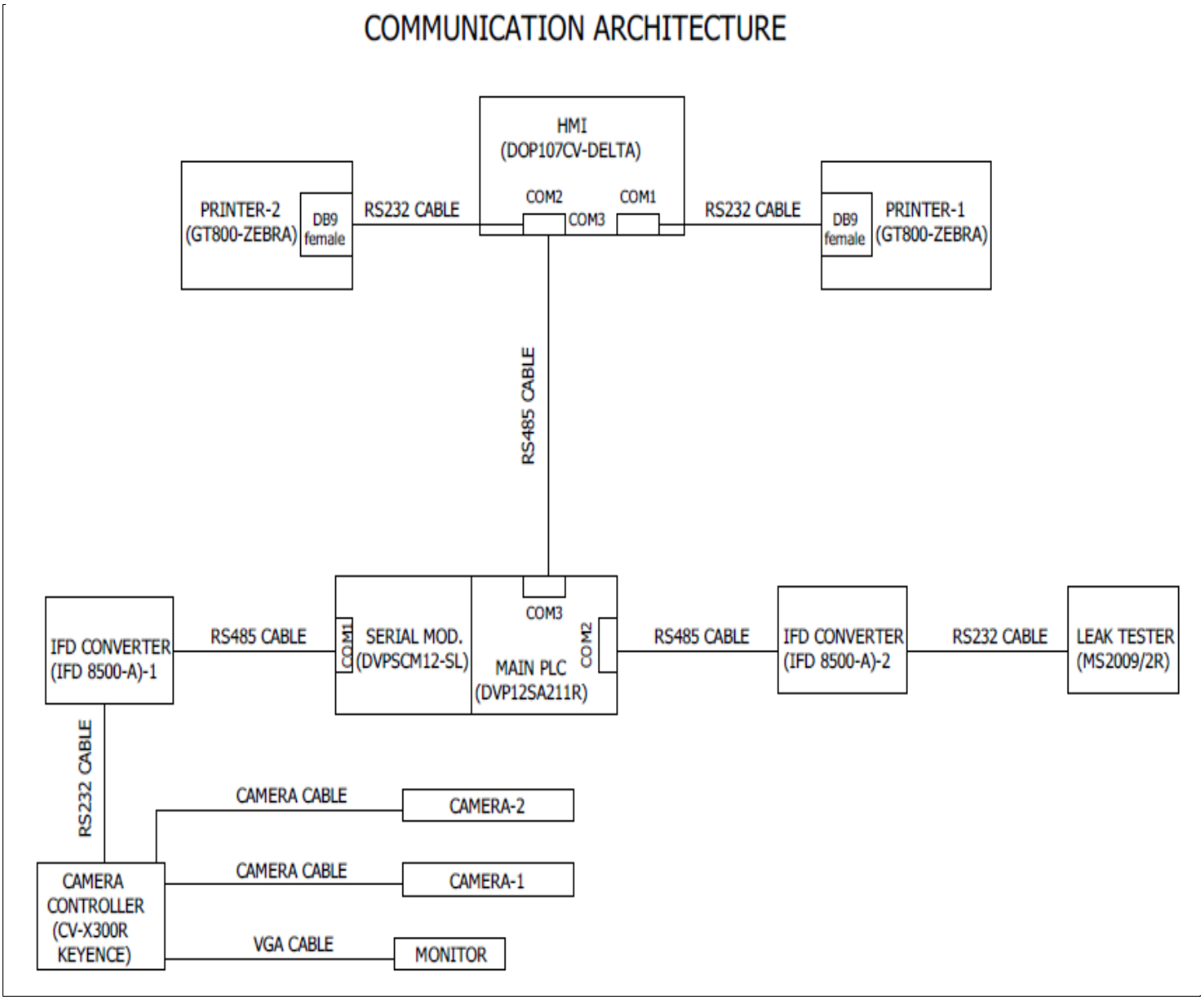
LH-PNEUMATIC LAYOUT



RH-PNEUMATIC LAYOUT



COMMUNICATION ARCHITECTURE



COMMUNICATION MODULE DVPSCM12 DETAILS

COMMUNICATION MODULE DVPSCM12 DETAILS

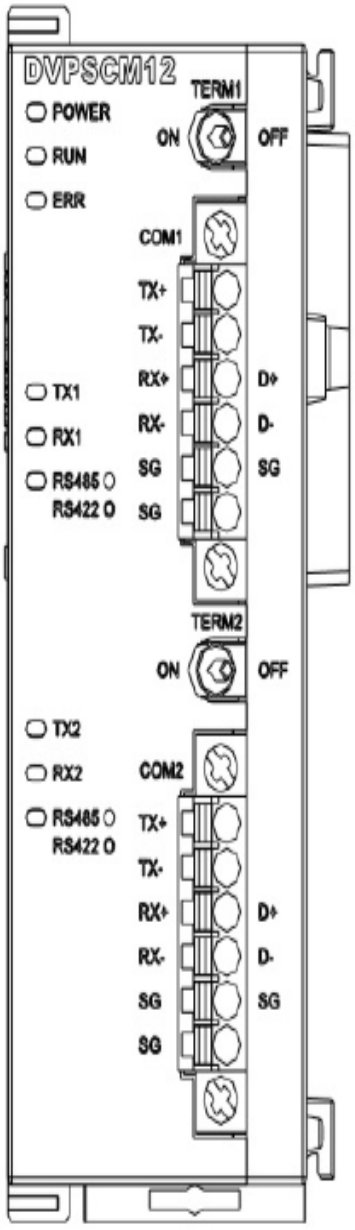
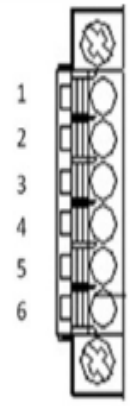
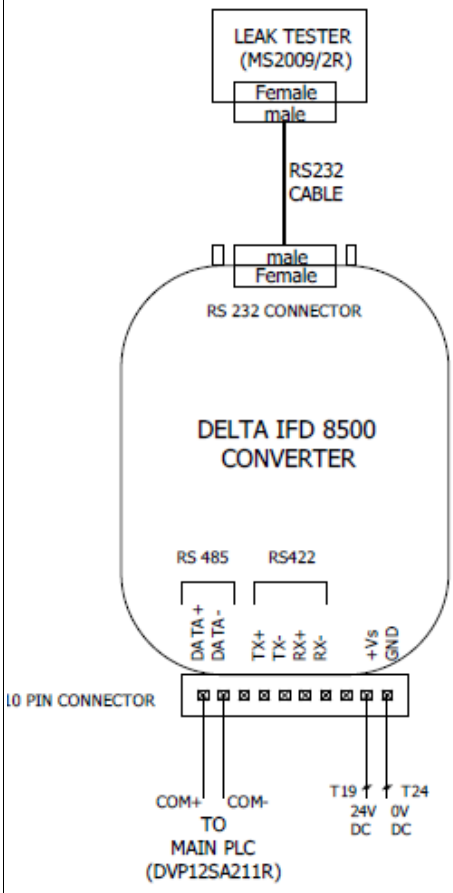
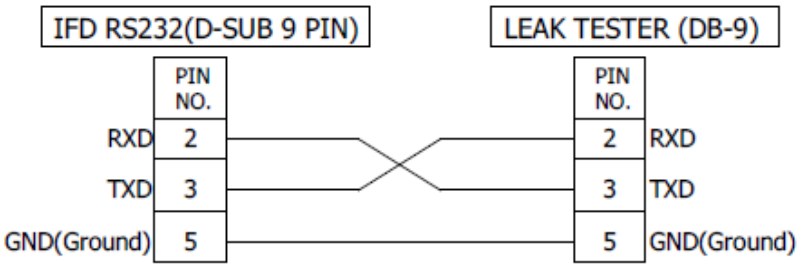


Diagram of RJ-45	Terminal no.	RS-485
	1	
	2	
	3	D+
	4	D-
	5	SG
	6	

RS485 TO RS232 (SERIAL IFD) CONVERTER CONNECTIONS

RS 485 TO RS 232(SERIAL IFD) CONVETER CONNECTIONS



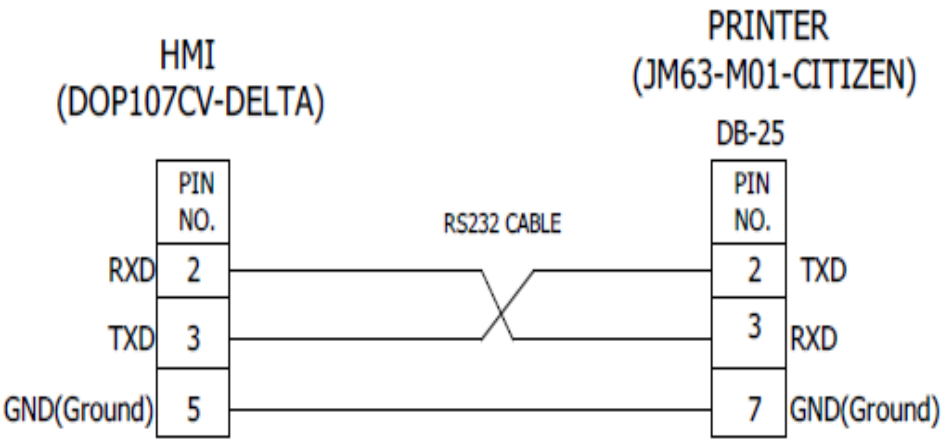


10 PIN CONNECTOR

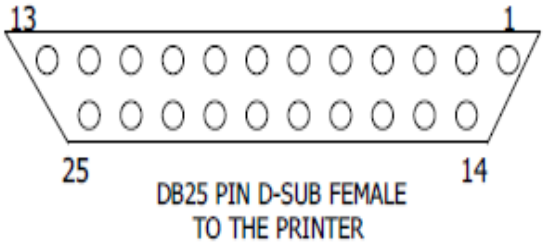
PIN	SIGNAL NAME
1	RS 485 DATA +
2	RS 485 DATA -
3	NC
4	RS 422 TX +
5	RS 422 TX -
6	RS 422 RX +
7	RS 422 RX -
8	NC
9	+VS (POWER)
10	GND(Ground)

HMI TO PRINTER CONNECTION

HMI TO PRINTER CONNECTION



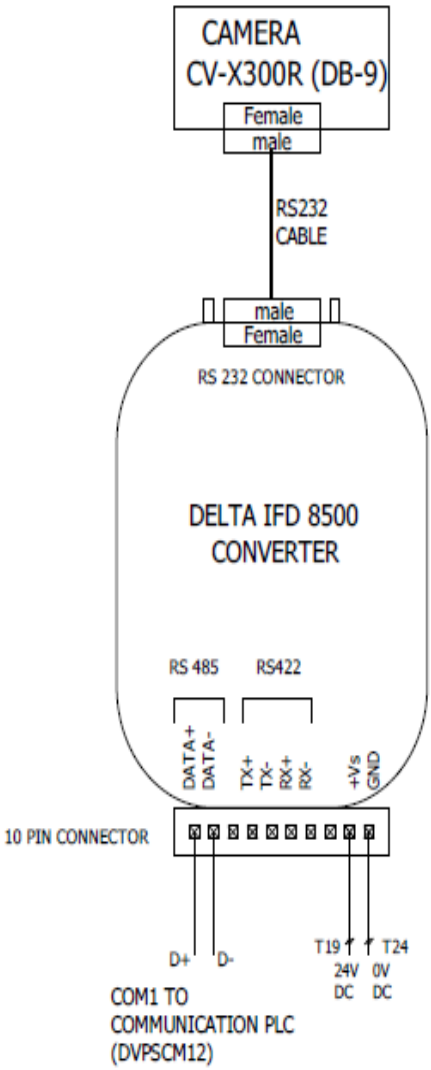
PIN		MODE-1	
	COM1	COM2	COM3
	RS-232	RS-232	RS-485
1	-	-	-
2	RXD	RXD	-
3	TXD	TXD	-
4	-	-	D+
5	GND		
6	-	-	-
7	RTS	RTS	-
8	CTS	CTS	-
9	-	-	D-

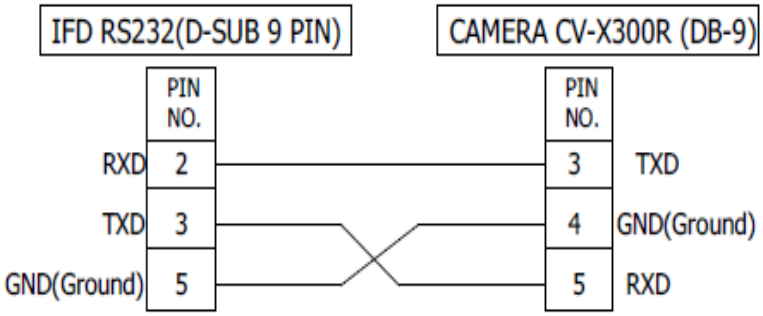


DB25 PIN D-SUB FEMALE
TO THE PRINTER

RS23 IFD TO RS232 (CAMERA-CV-X300R) CONVERTER CONNECTIONS

RS 232 IFD TO RS 232(CAMERA-CV-X300R) CONVETER CONNECTIONS

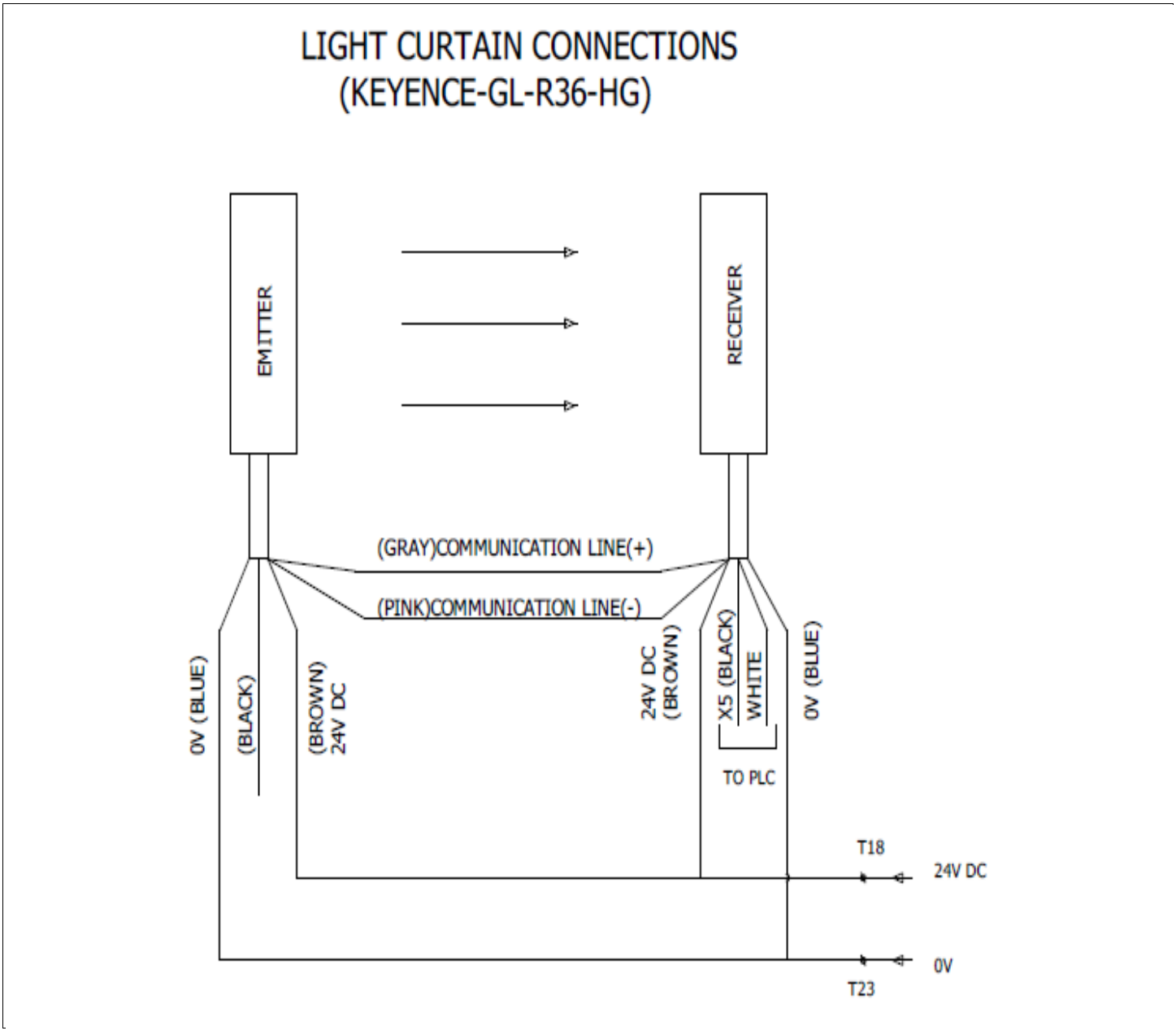




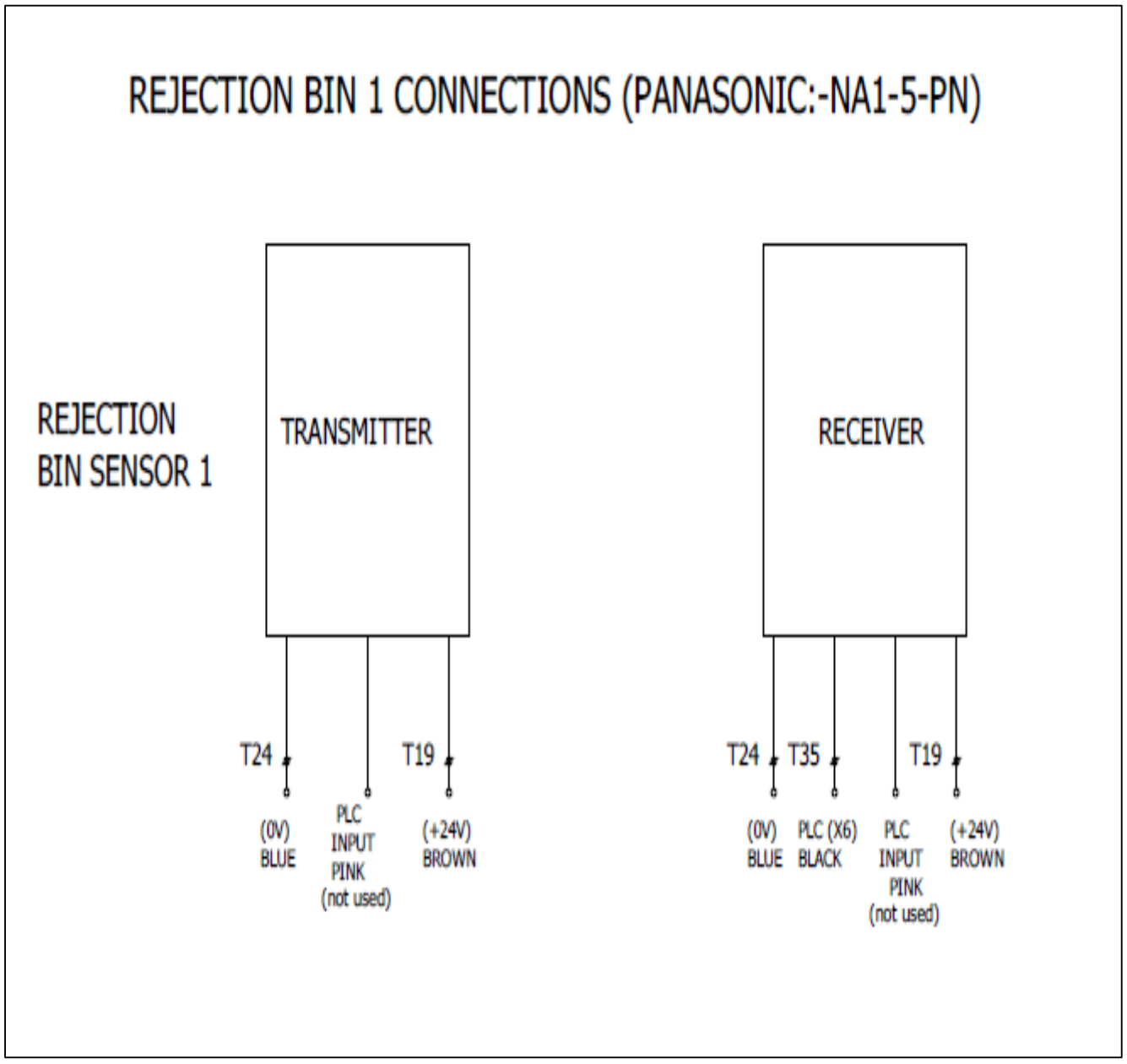
10 PIN CONNECTOR

PIN	SIGNAL NAME
1	RS 485 DATA +
2	RS 485 DATA -
3	NC
4	RS 422 TX +
5	RS 422 TX -
6	RS 422 RX +
7	RS 422 RX -
8	NC
9	+VS (POWER)
10	GND(Ground)

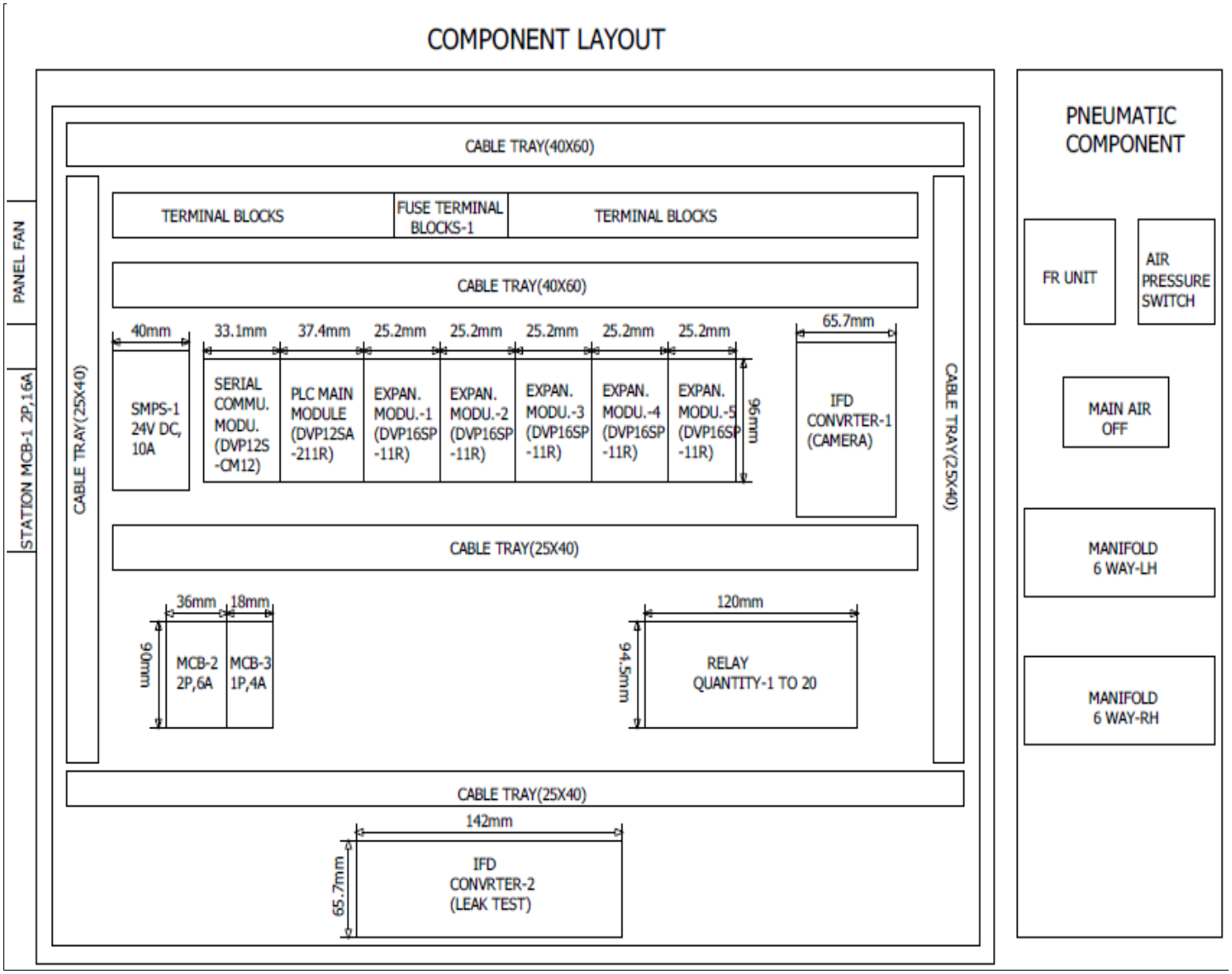
LIGHT CURTAIN CONNECTIONS



REJECTION BIN CONNECTION

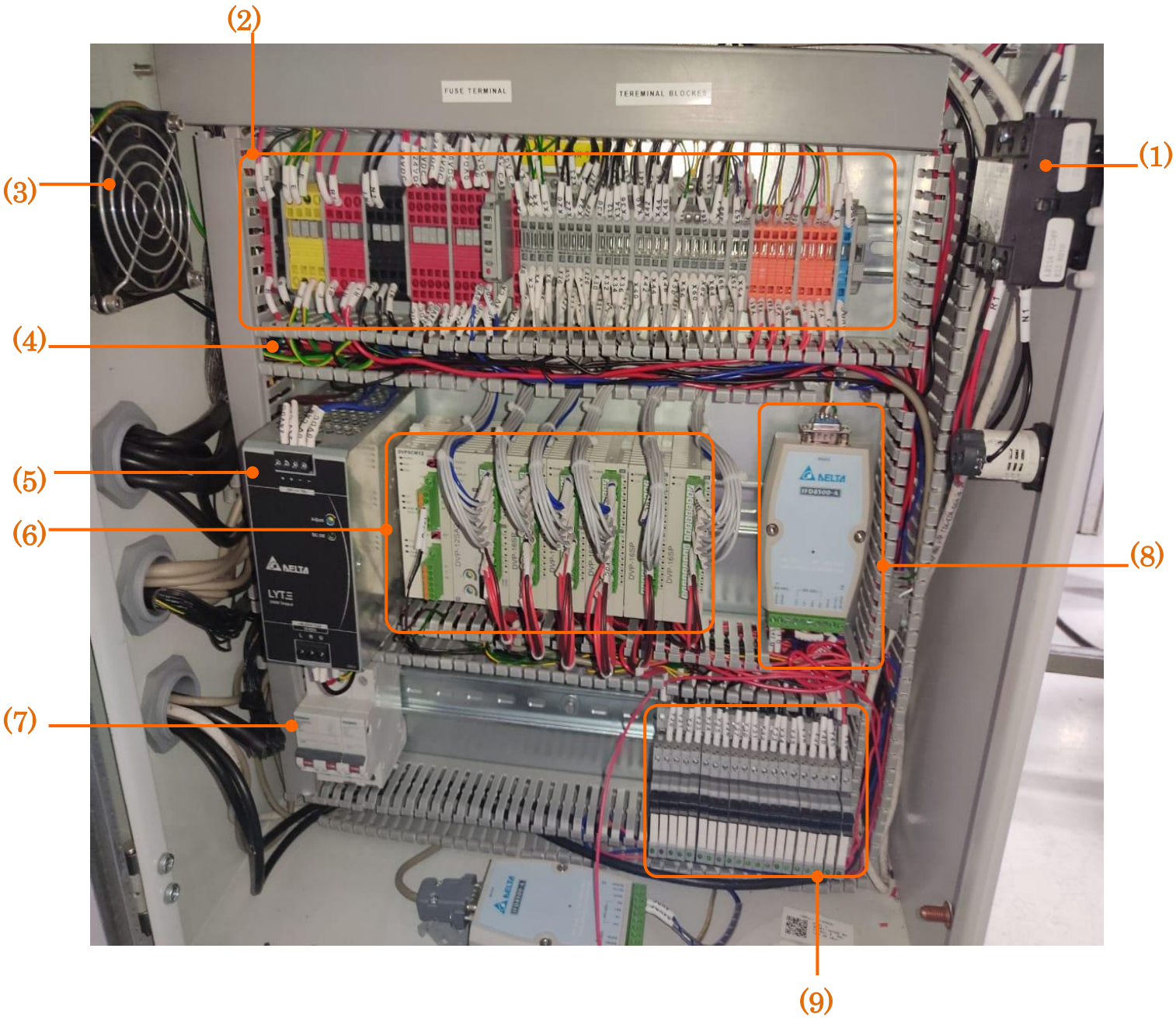


COMPONENT LAYOUT



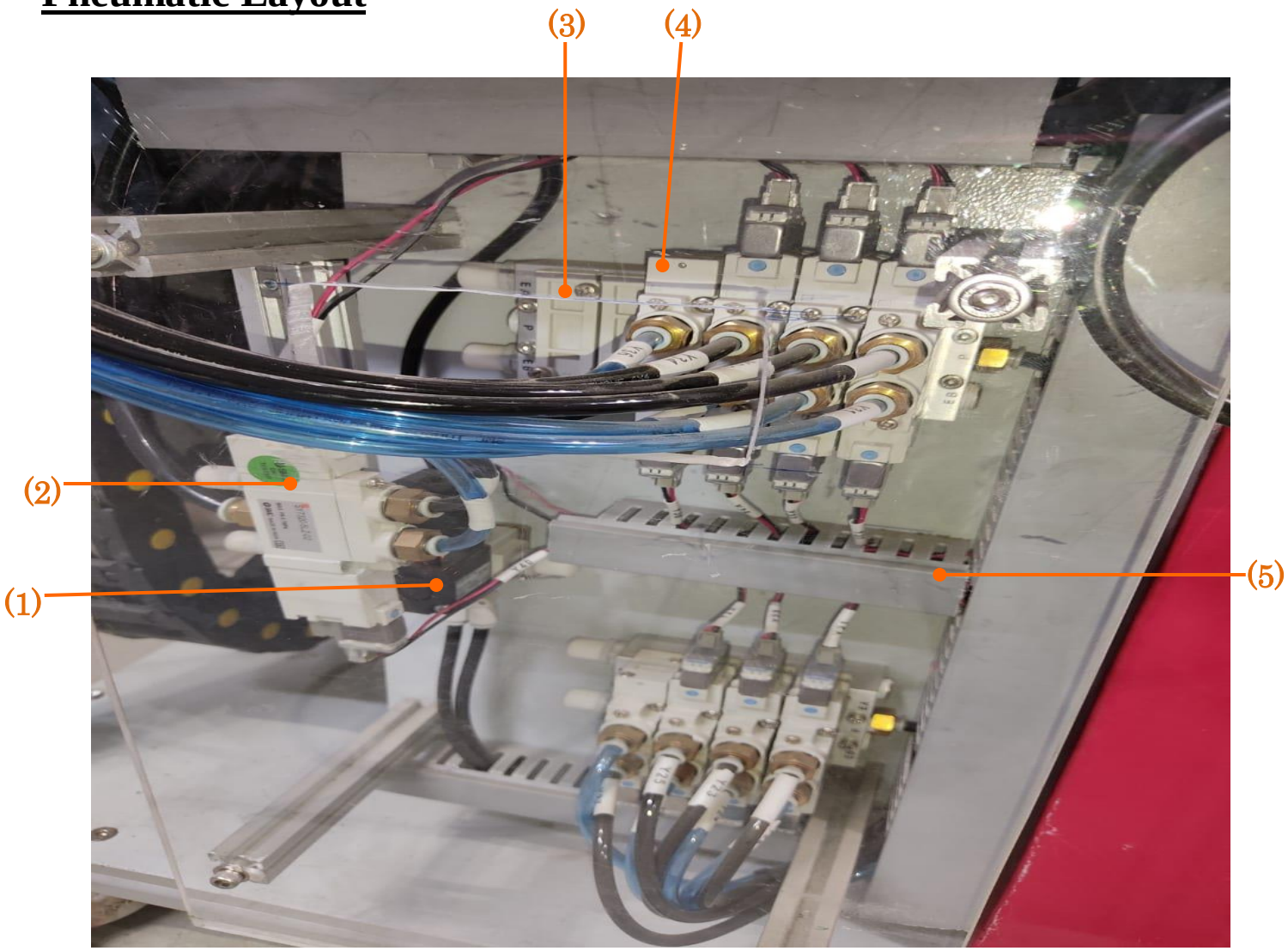
Component Layout

Internal Panel Layout and Pneumatic Plate Details



No.	Component Name	Description
1	On /off switch	To on or off the machine power.
2	Terminal Blocks	To connect external devices.
3	Panel fan	To heat dissipation of panel.
4	Cable Tray	To Route Internal Wires.
5	SMPS (24 V, 2.5 A)	Used for AC to DC converter.
6	PLC	To input output controller.
7	MCB	To protection against short circuit and overloading current.
8	IFD	To convert RS485 TO RS232 conversion.
9	1-channel relays	To connect solenoid valves.

Pneumatic Layout



No.	Component Name	Description
1	Air Off Solenoid Valve	To on /off air
2	FR Unit	Air filter & regulator unit.
3	manifold	Suitable mounting of solenoid valve.
4	Double acting solenoid valve	For cylinder operation.
5	Cable Tray	To Route Internal Wires.

BILL OF MATERIAL:

YFG FCL LEAK TEST BOM					
ELECTRICAL PANEL MATERIAL					
SN	**COMPONENT NAME**	**DESCRIPTION**	**PART NO.**	**MAKE**	**QTY.**
1	Control Panel	500X500X210	MAS0505021R5	Nventhoffman	1
2	PLC	8DI,4DO	DVP12SA2	Delta	1
3	Expansion module	8DI,8DO	DVP16SP11R	Delta	5
4	Expansion module	RS485	DVPSCM12-SL	Delta	1
5	RS 485 TO 232	RS 485 to 232 Converter	IFD8500-A	Delta	2
6	HMI		DOP-107CV	Delta	1
7	SMPS	24 VDC,10A	S8VK-C24024	Omron	1
8	Push Button (Green)	Green Mashrum Type	2AM3	Teknic	2
9	Reset Button(red)	Red Flush Type	2AM4	Teknic	1
10	Emergency Stop	Latched	2AML4	Teknic	1
11	NO Element	24 VDC	S1	Teknic	3
12	NC Element	24 VDC	S2	Teknic	1
13	PB Box	2Way	50MM	Teknic	2
14	Indication Lamp (Green)	230 VAC	3PLBR-3L	Teknic	1
15	Relay	24V,6A	SOCY1-24D	Connectwell	19
16	ON/Off Switch	2P,16A,230VAc	61002SAB13TDYR	L&t	1
17	MCB (SMPS Input)	2P,6A	5SL62067RC	Siemens	1
18	MCB (Pannel)	2P,16A	5SL42167RC	Siemens	1
19	MCB (24VDC)	1P,4A	5SL61047RC	Siemens	1
20	Tower Lamp 3-Tier	Steady,24 VDC	MG-302B-RYG	Patellite	1
21	Fuse terminal	4A		Wago	5

22	Electric indo	24 Pin Regular, male female with housing		Indo	2
23	Panel fan	230VAC,18W,0.10A,2650RPM	12A230HSAC	Hicool	1
24	LED Tube Light	230VAC	LP-201-15W	Lord's	2
25	Extension Board & Switch	6A(Monitor-1,Printer-2,Leak tester-1 spare-2)		Legrand	2
26	Station Fan	230VAC,22W,0.10A,2650RPM	12A230HSAC	Hicool	1
27	Rejection bin sensor		NA1-5-PN,UAN 15PN	Panasonic	1
28	Industrial socket	230v,3P,16A		Legrand	1
29	Camera controller	Vision system controller	CVX-300R	Keyence	1
30	Camera	Vision system camera	CA-H048MX	Keyence	2
31	Camera cable				2
32	Light curtain	Transmitter	GL-R40HG-T	Keyence	1
		Receiver	GL-R40HG-R	Keyence	1
33	Monitor +VGA cable			Keyence	1
34	Printer		GT800	Zebra	2
35	Leak tester	230VAC,80W, LEAK DETECTOR	MS2009/2R	E.C.B.S.r.l.	1

YFG FCL LEAK TEST					
Sr No	**Component Name**	**Description**	**Part No.**	**Make**	**Qty.**
1	Solenoid Valve	Double Acting	SY7227-5LZ-02	SMC	7
		Single Acting	SY7120-5LZ-02	SMC	2
2	Read switch	24VDC, PNP		FESTO	18
3	Air Off Solenoid Valve	0.7MPa	VT307-5DZ1-02	SMC	1
4	digital pressure switch	Pressure Display	ISE30AF-01-P-LA2	SMC	1
5	FRL unit	1/4"	AW20-02BG1-B	SMC	1
6	Quick Coupling connector 10 pin	Male+female	KDM10S-06	SMC	2
7	Manifold	6 Way manifold	SS5Y7-20-06	SMC	2

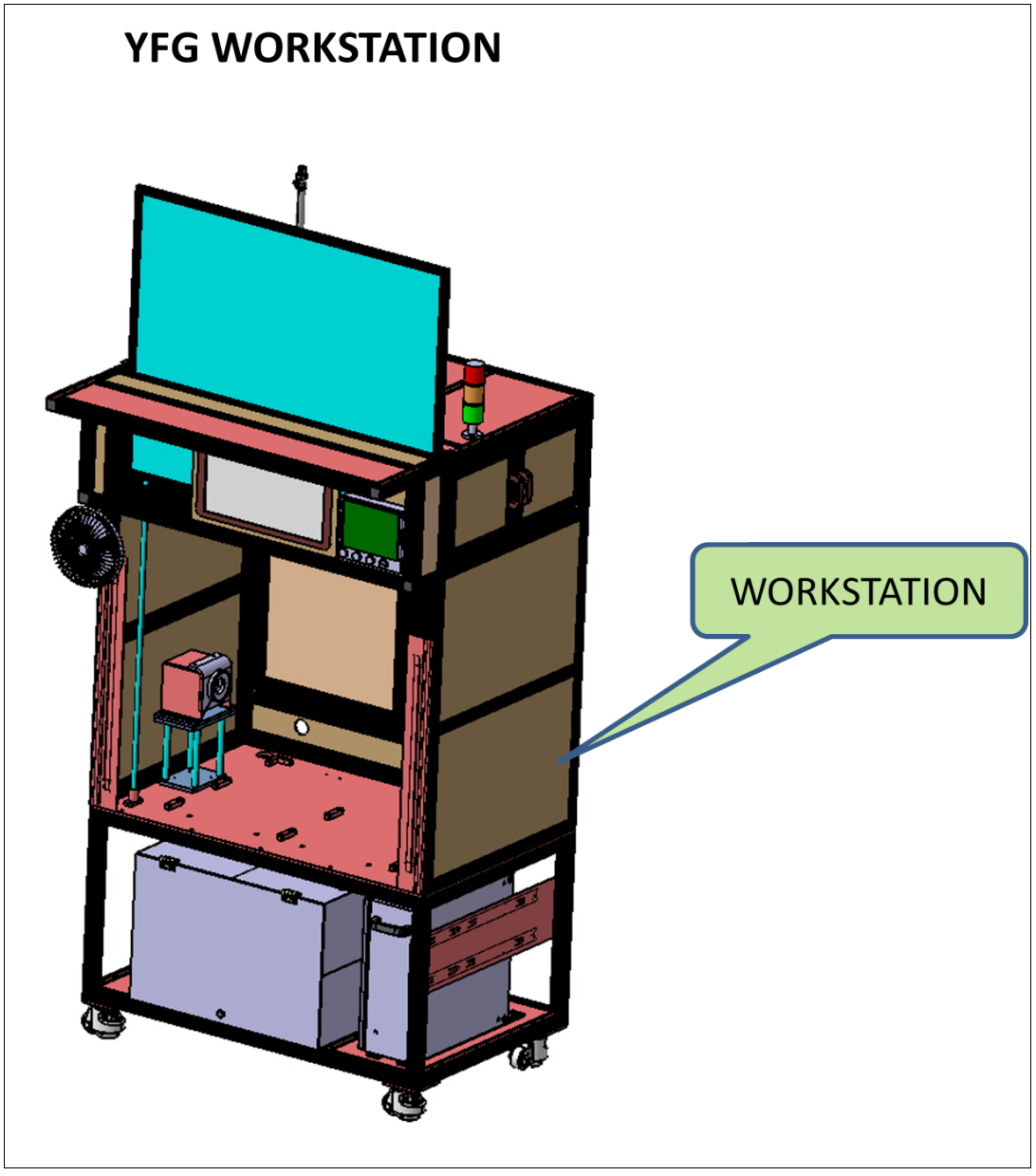
I/O LIST:

YFG FCL LEAK TEST					
MODULES	I/P ADDR	INPUTS	O/P ADDR	OUTPUTS	RELAY
DVP 12SA2	X0	START PB 1	Y0	PASS LAMP	
	X1	START PB 2	Y1	BUZZER	
	X2	EMB PB	Y2	PROCESS LAMP	
	X3	RESET	Y3	ERROR LAMP	
	X4	AIR PRESSURE SWITCH	NA	NA	
	X5	LIGHT CURTAIN	NA	NA	
	X6	SPARE	NA	NA	
	X7	SPARE	NA	NA	
DVP 16SP11R-1	X20	LH PART PRESENT SENSOR 1	Y20	AIR OFF SOL	R3
	X21	LH PART PRESENT SENSOR 2	Y21	LH CLMAP CYL FWD SOL	R4
	X22	LH PART PRESENT SENSOR 3	Y22	LH CLMAP CYL RET SOL	R5
	X23	LH PART PRESENT SENSOR 4	Y23	LH SEAL CYL FWD SOL	R6
	X24	LH CLAMP CYL FWD RD	Y24	LH SEAL CYL RET SOL	R7
	X25	LH CLAMP CYL RET RD	Y25	LH FILING CYL FWD SOL	R8
	X26	LH SEAL CYL FWD RD	Y26	LH FILING CYL RET SOL	R9
	X27	LH SEAL CYL RET RD	Y27	LH DOT PUNCH FWD/RET SOL	R10
DVP 16SP11R-2	X30	LH FILING CYL FWD RD	Y30	RH CLAMP CYL FWD SOL	R11
	X31	LH FILING CYL RET RD	Y31	RH CLAMP CYL RET SOL	R12
	X32	LH DOT PUNCH FWD RD	Y32	RH SEAL CYL FWD SOL	R13
	X33	LH DOT PUNCH RET RD	Y33	RH SEAL CYL RET SOL	R14
	X34	RH PART PRESENT SENSOR 1	Y34	RH FILING CYL FWD SOL	R15
	X35	RH PART PRESENT SENSOR 2	Y35	RH FILING CYL RET SOL	R16
	X36	RH PART PRESENT SENSOR 3	Y36	RH DOT PUNCH FWD/RET SOL	R17
	X37	RH PART PRESENT SENSOR 4	Y37	SHUTTER CYL FWD SOL	R18
DVP 16SP11R-3	X40	RH CLAMP CYL FWD RD	Y40	SHUTTER CYL RET SOL	R19
	X41	RH CLAMP CYL RET RD	Y41	LH LEAK TEST START	
	X42	RH SEAL CYL FWD RD	Y42	LH LEAK TEST STOP	
	X43	RH SEAL CYL RET RD	Y43	RH LEAK TEST START	
	X44	RH FILING CYL FWD RD	Y44	RH LEAK TEST STOP	
	X45	RH FILING CYL RET RD	Y45	PANEL FAN	R20
	X46	RH DOT PUNCH FWD RD	Y46	SPARE	
	X47	RH DOT PUNCH RET RD	Y47	SPARE	
DVP 16SP11R-4	X50	SHUTTER CYL FWD RD	Y50	LEAK PROGRAM SEL-1	
	X51	SHUTTER CYL RET RD	Y51	LEAK PROGRAM SEL-2	
	X52	LH LEAK TEST PASS	Y52	LEAK PROGRAM SEL-3	
	X53	LH LEAK TEST FAIL	Y53	LEAK PROGRAM SEL-4	

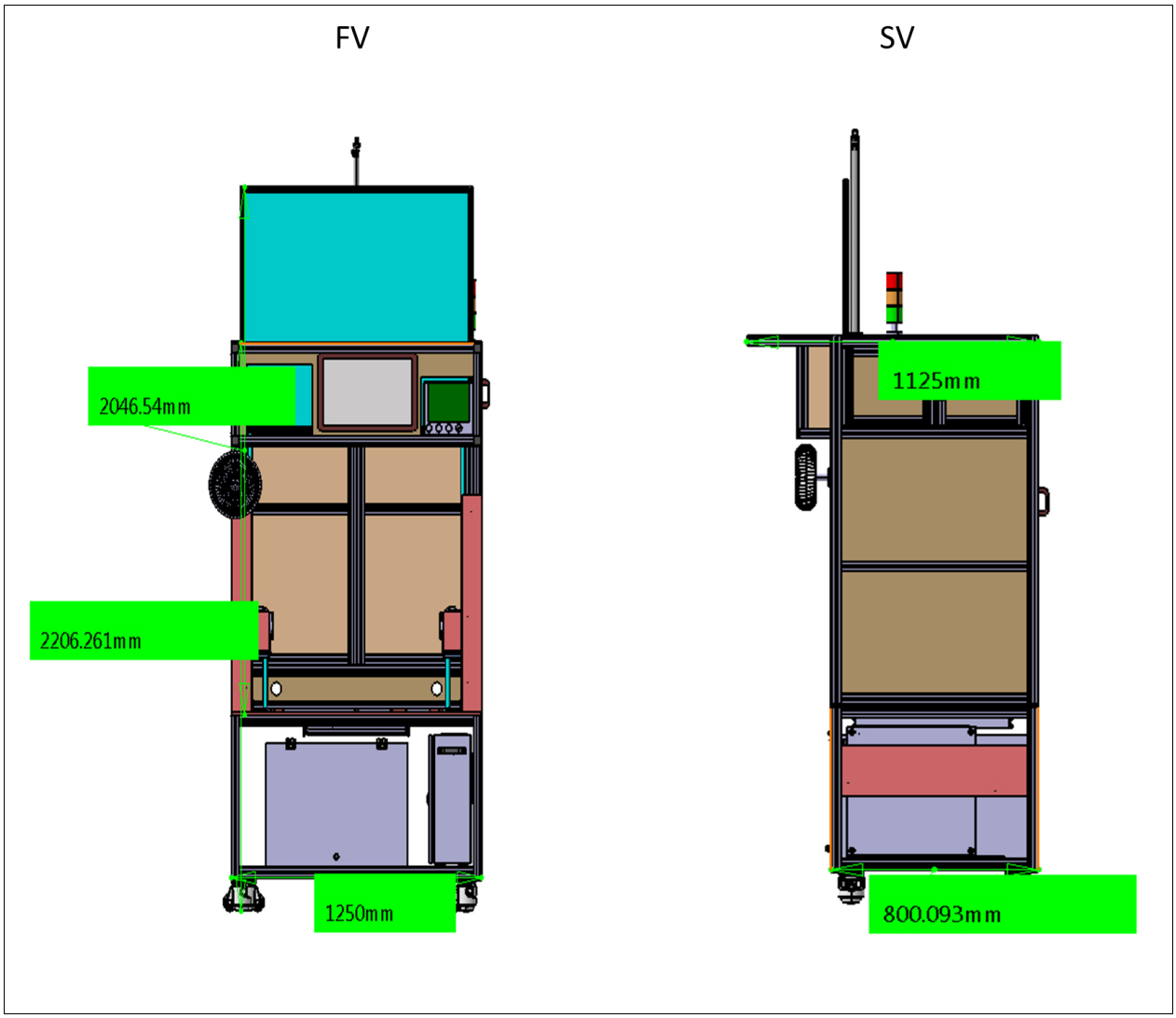
	X54	RH LEAK TEST PASS	Y54	SPARE	
	X55	RH LEAK TEST FAIL	Y55	SPARE	
	X56	REJECTION BIN SENSOR-1	Y56	SPARE	
	X57	REJECTION BIN SENSOR-2	Y57	SPARE	
DVP 16SP11R-5	X60	REJECTION BIN SENSOR-3	Y60	SPARE Station tubelight-1 (on station side)	R21
	X61	REJECTION BIN SENSOR-4	Y61	Station tubelight-2 (on fixture RH)	R2
	X62	LH VARIENT SEL-1	Y62	Station tubelight-2 (on fixture LH)	R1
	X63	LH VARIENT SEL-2	Y63	SPARE	
	X64	LH VARIENT SEL-3	Y64	SPARE	
	X65	RH VARIENT SEL-1	Y65	SPARE	
	X66	RH VARIENT SEL-2	Y66	SPARE	
	X67	RH VARIENT SEL-3	Y67	SPARE	
COMMUNICATION	PLC -HMI	PLC COM3 -HMI COM3			
	PLC -LEAK TEST	PLC COM2- RS232 TO 485 CONVERTER -LEAK TEST			
	PLC - CAMERA	DVPSCM12-SL RS232 TO 485 CONVERTER			
	HMI - PRINTER 1	HMI COM1			
	HMI - PRINTER 2	HMI COM2			

WORKSTATION DETAILS:

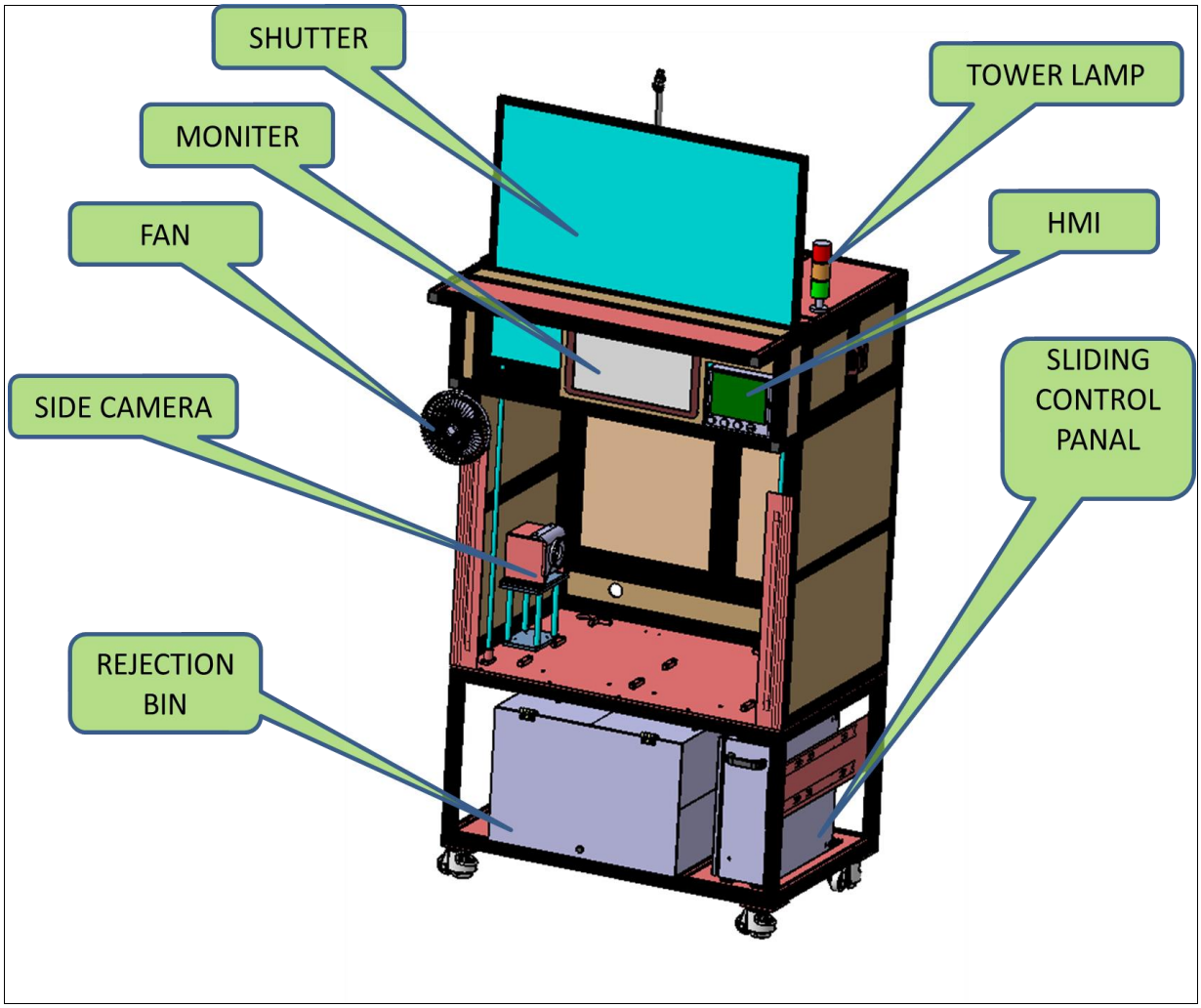
COMPONENT DESCRIPTION



OVERALL DIMENSION

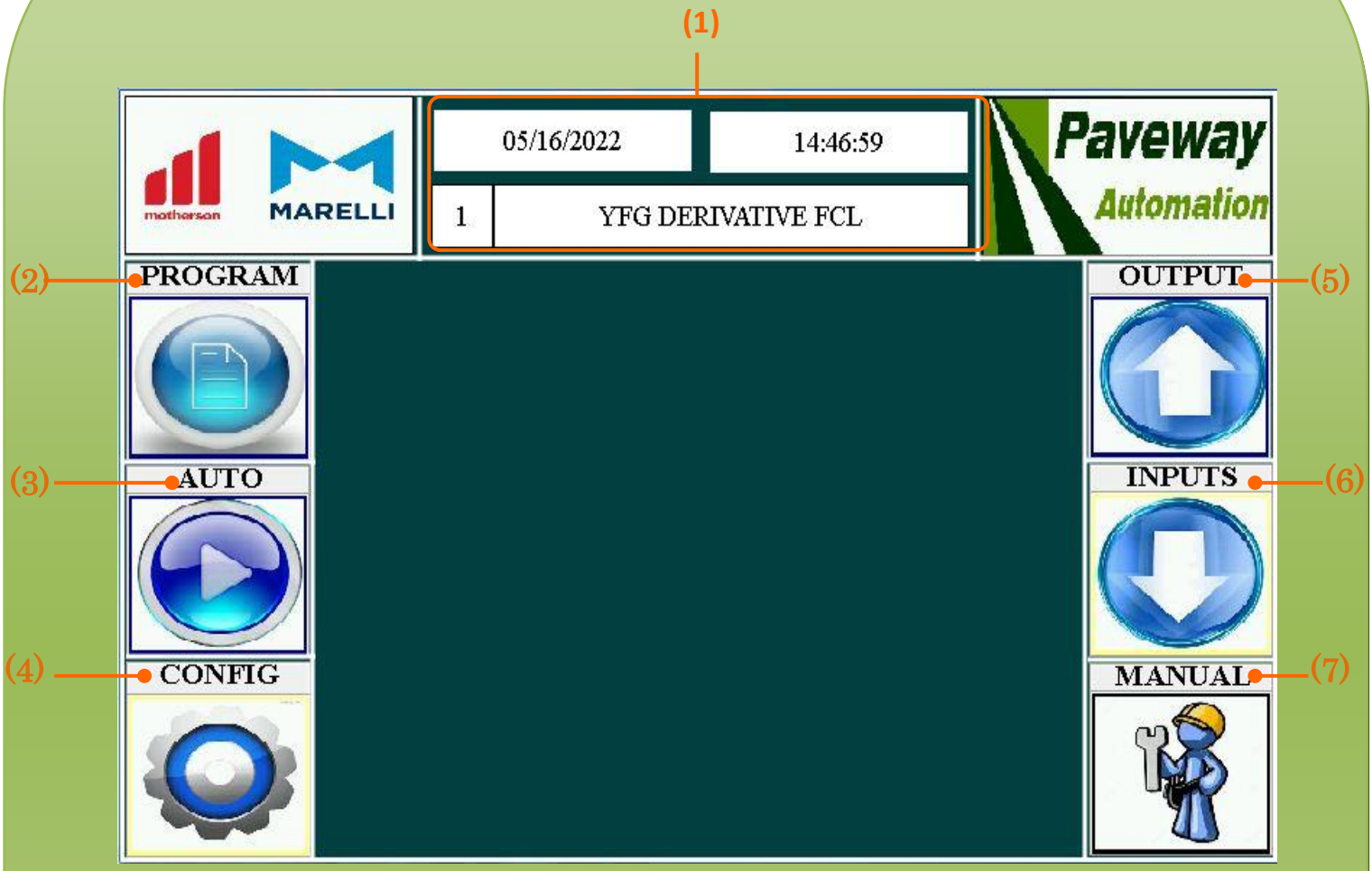


FIXTURE DETAILS



HMI SCREEN DETAILS:

HOME SCREEN



1.Program name, Date & Time display.

2.Program monitering screen button .

3.Auto mode screen navigation button.

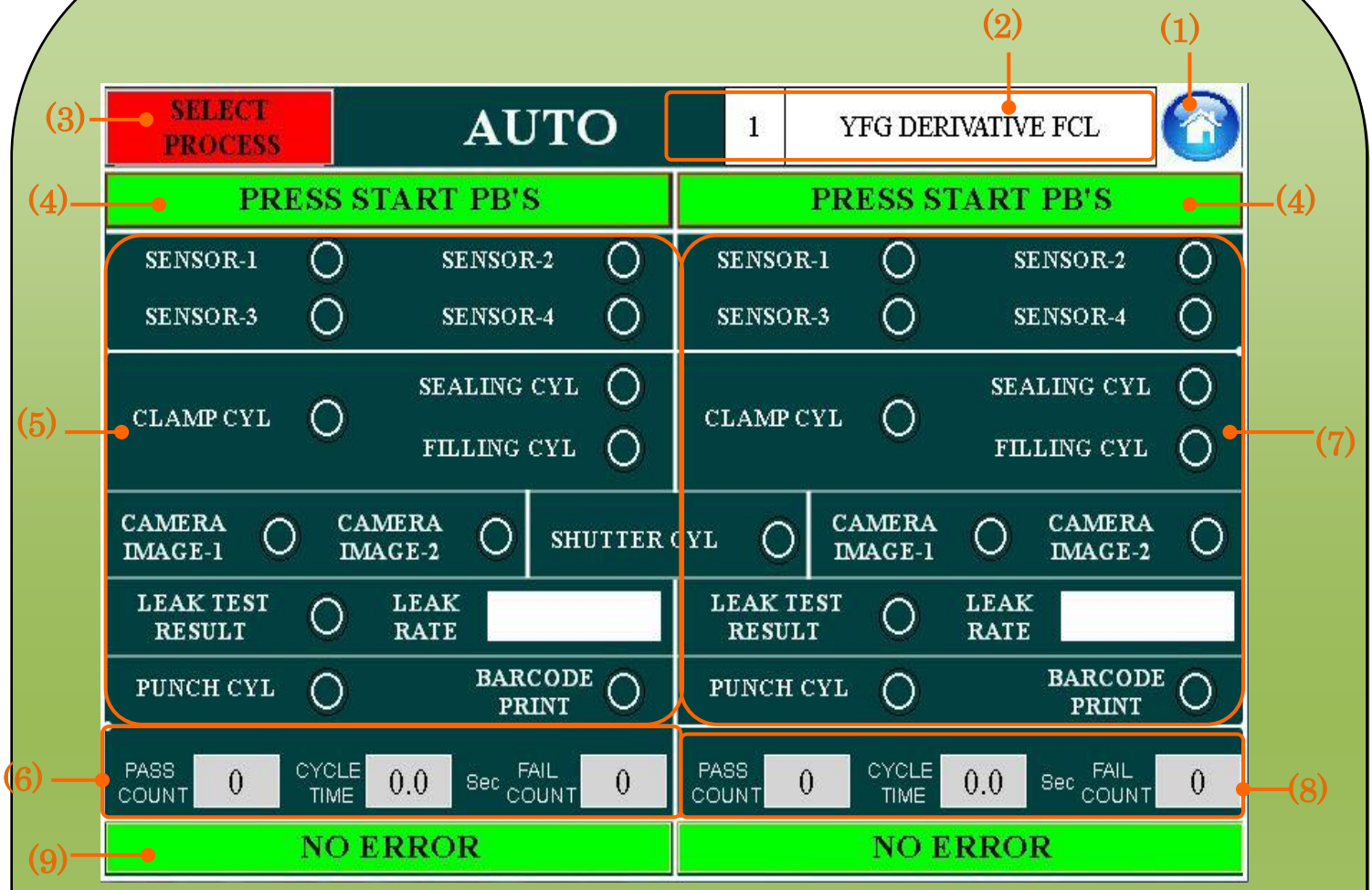
4.Configuration setting navigation button.

5.Output navigation button.

6.Input navigation button.

7. Manual navigation button.

AUTO MODE SCREEN



1.Home screen navigation button.

2. Program name & number.

3.Select process button.

4. Status message display.

5. LH sensors status display.

- LH clamp,sealing,filling cylinder status display.
- Camera image-1 and camera image-2 status display.
- Shutter cylinder status display.

- Leak test result and leak rate status display.
- Punch cylinder and barcode print status display.

6. LH side pass count, cycle time per second and fail count status display.

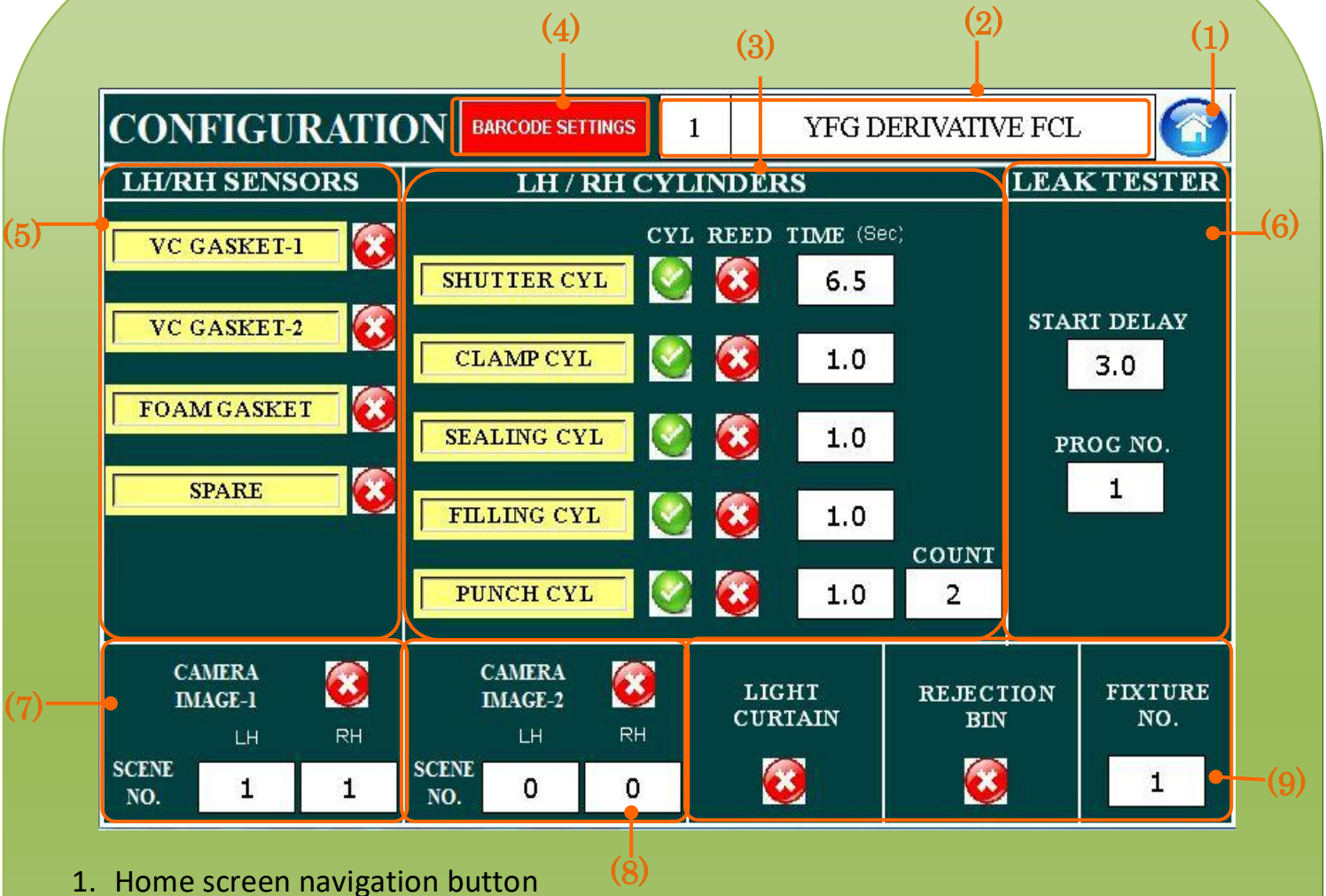
7. RH sensors status display.

- RH clamp,sealing,filling cylinder status display.
- Camera image-1 and camera image-2 display.
- Shutter cylinder status display.
- Leak test result and leak rate status display.
- Punch cylinder and barcode print status display.

8. RH side pass count, cycle time per second and fail count status display.

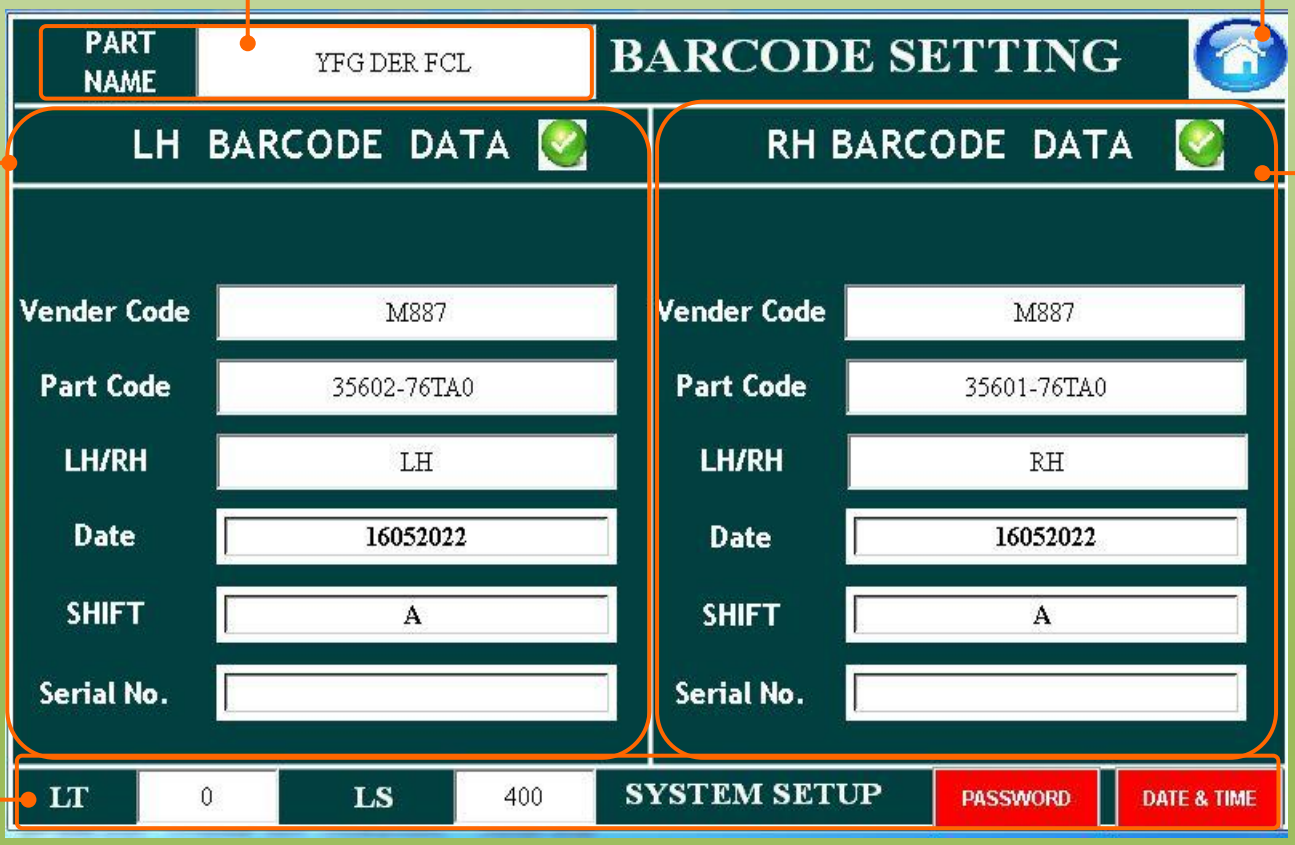
9. Error message display.

CONFIGURATION SCREEN



1. Home screen navigation button
2. Program name & number display.
3. LH/RH cylinders select/unselect option and time per second display.
4. Barcode setting navigation button.
5. LH/RH sensor select/unselect option.
6. Leak tester :-start delay display and program number display.
7. Camera image-1 select/unselect option and LH/RH scene number display.
8. Camera image-2 select/unselect option and LH/RH scene number display.
9. Light curtain and rejection bin select/unselect option and fixture number display.

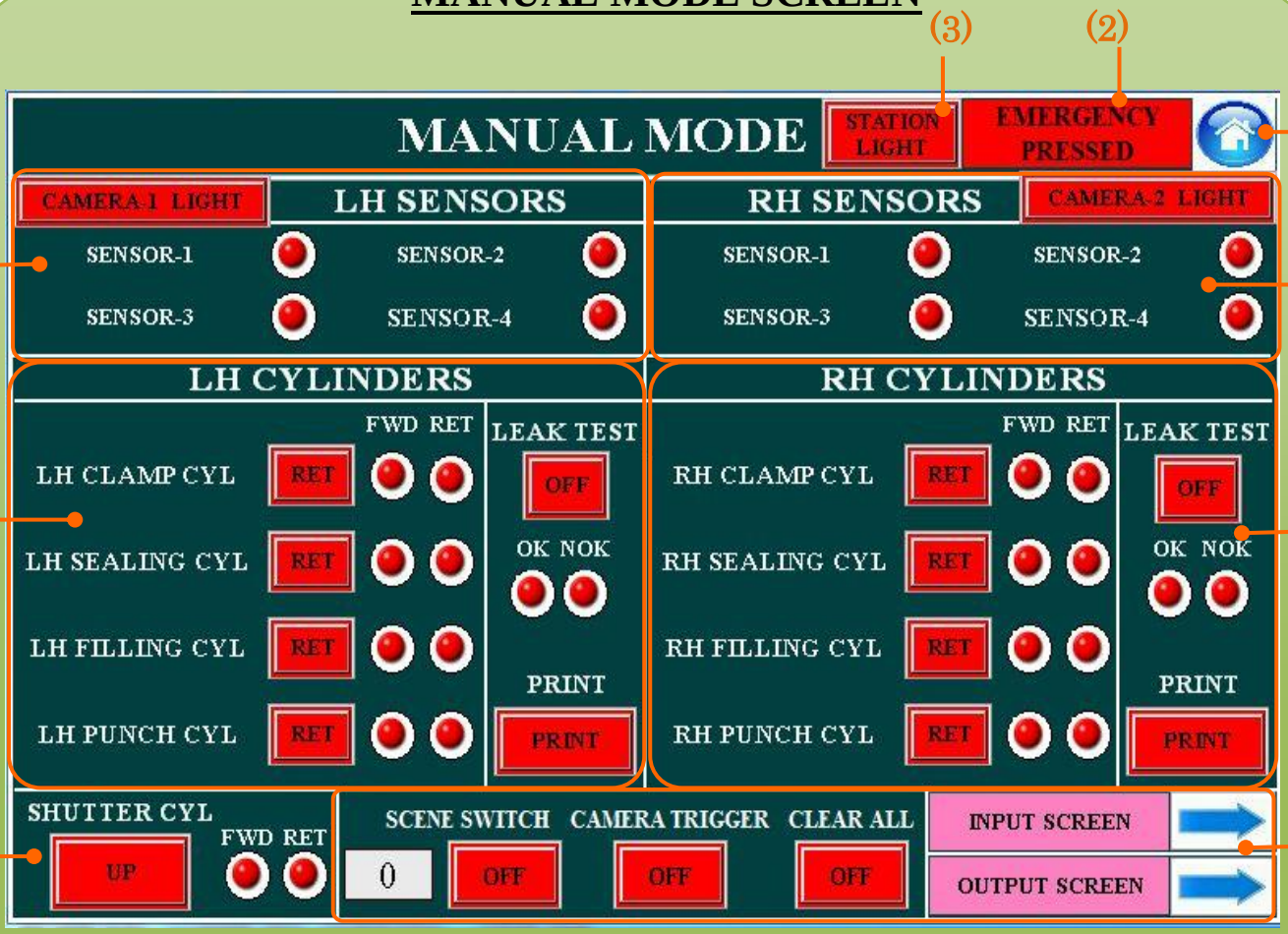
BARCODE SETTING SCREEN



1. Home screen navigation button
2. Part name display.
3. LH side barcode data setting.
4. RH side barcode data setting.
5. LT and LS value display.

- System setup: -
- Password and date time setting option.

MANUAL MODE SCREEN



1. Home screen navigation button.
2. Emergency pressed status display.
3. Station light status display.
4. LH camera-1 light manual operation status display.
   - LH sensors manual operation status display.
5. RH camera-2 light manual operation status display.
   - RH sensors manual operation status display.
6. LH clamp, sealing, filling, punch cylinder manually press fwd/ret status display.
   - Leak test on/off manual operation status display.

- LH side print navigation button.

7. RH clamp, sealing, filling, punch cylinder manually press fwd/ret status display.

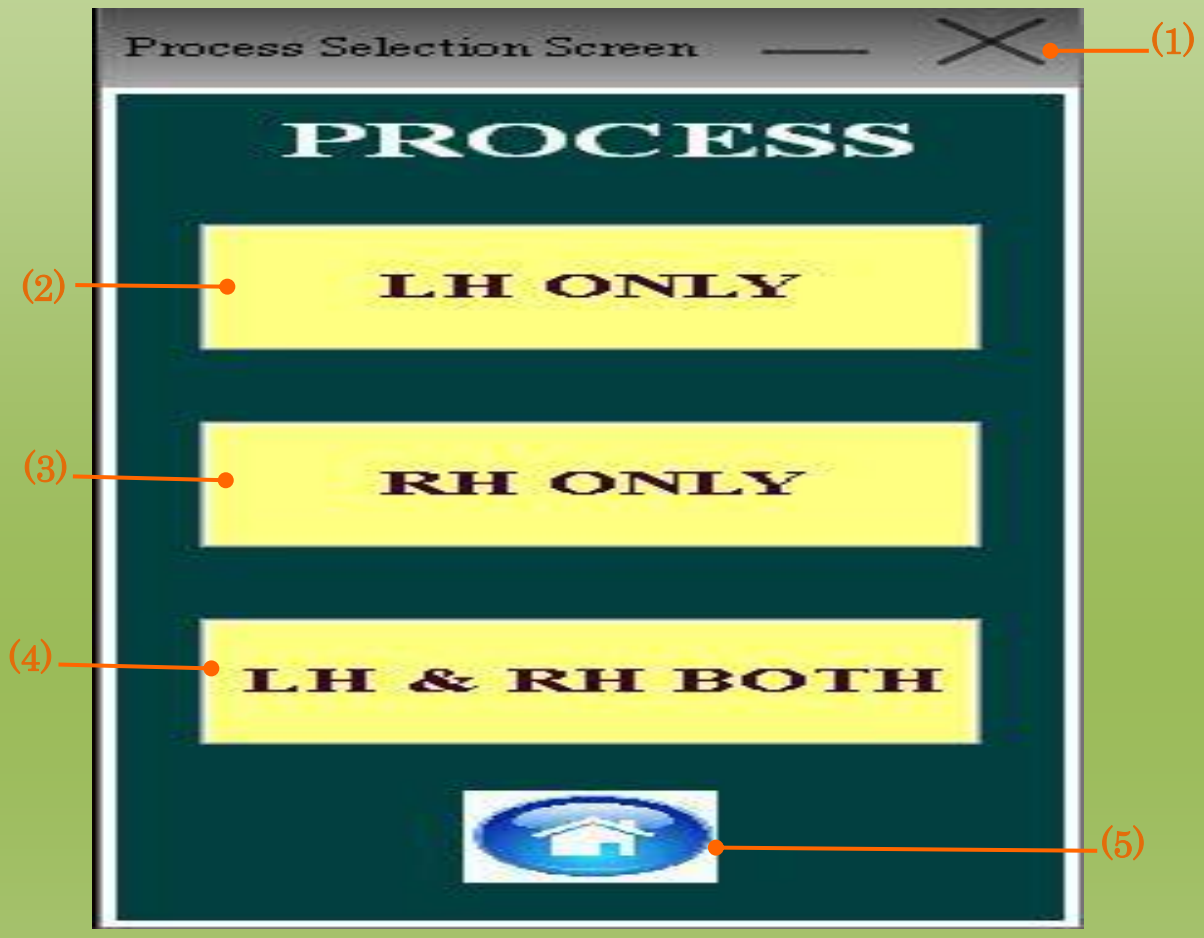
- Leak test on/off manual operation status display.
- RH side print navigation button.

8. Shutter cylinder manually press fwd/ret operation status display.

9. Scene switch, camera trigger, clear all manually press on/off operation status display.

- Input screen next page navigation button.
- Output screen next page navigation button.

PROCESS SELECTION SCREEN



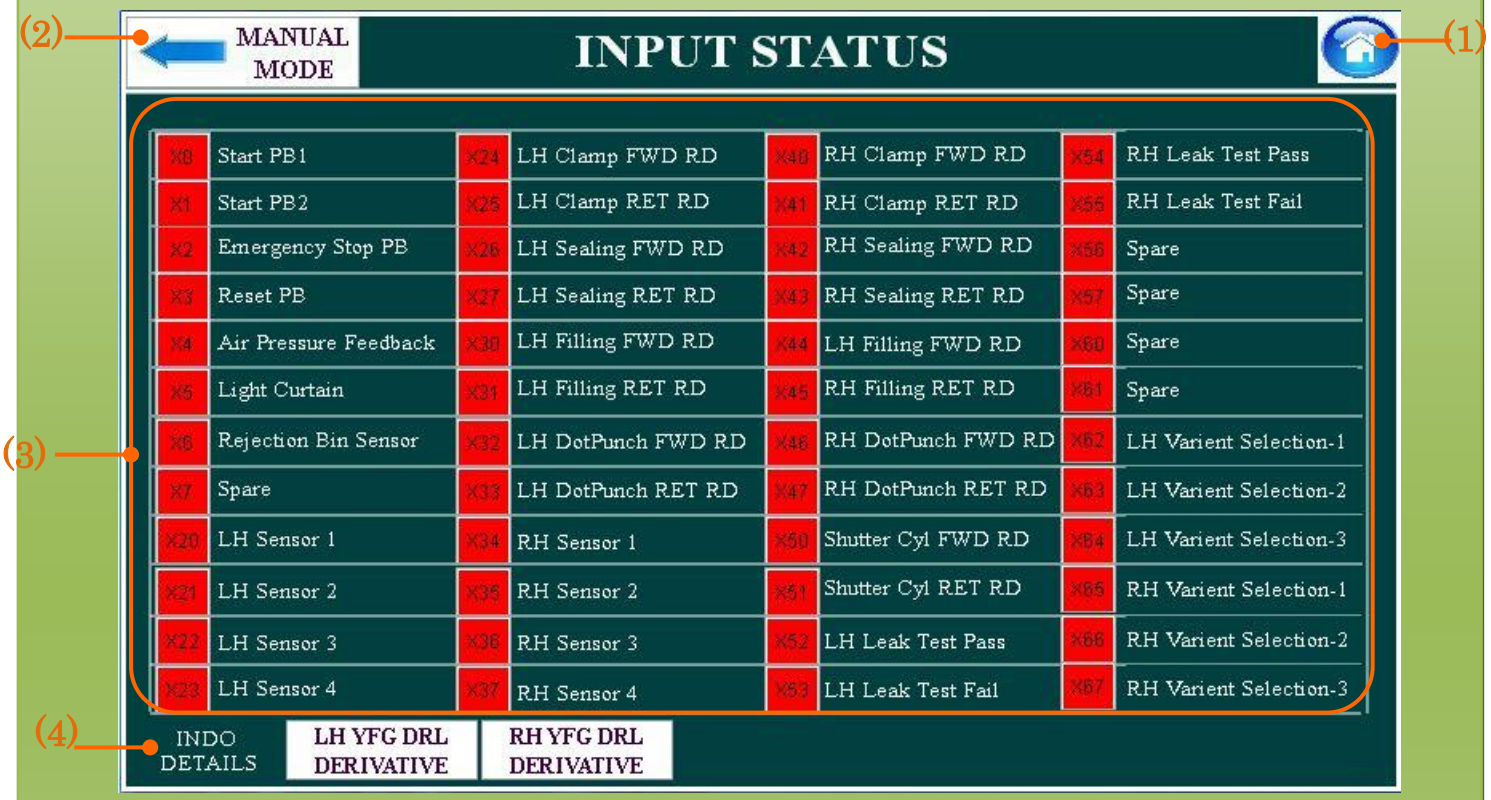
1. Minimize or maximize screen navigation option.
2. LH only select option.
3. RH only select option.
4. LH and RH both select option.
5. Home screen navigation button.

PROGRAM SELECTION SCREEN



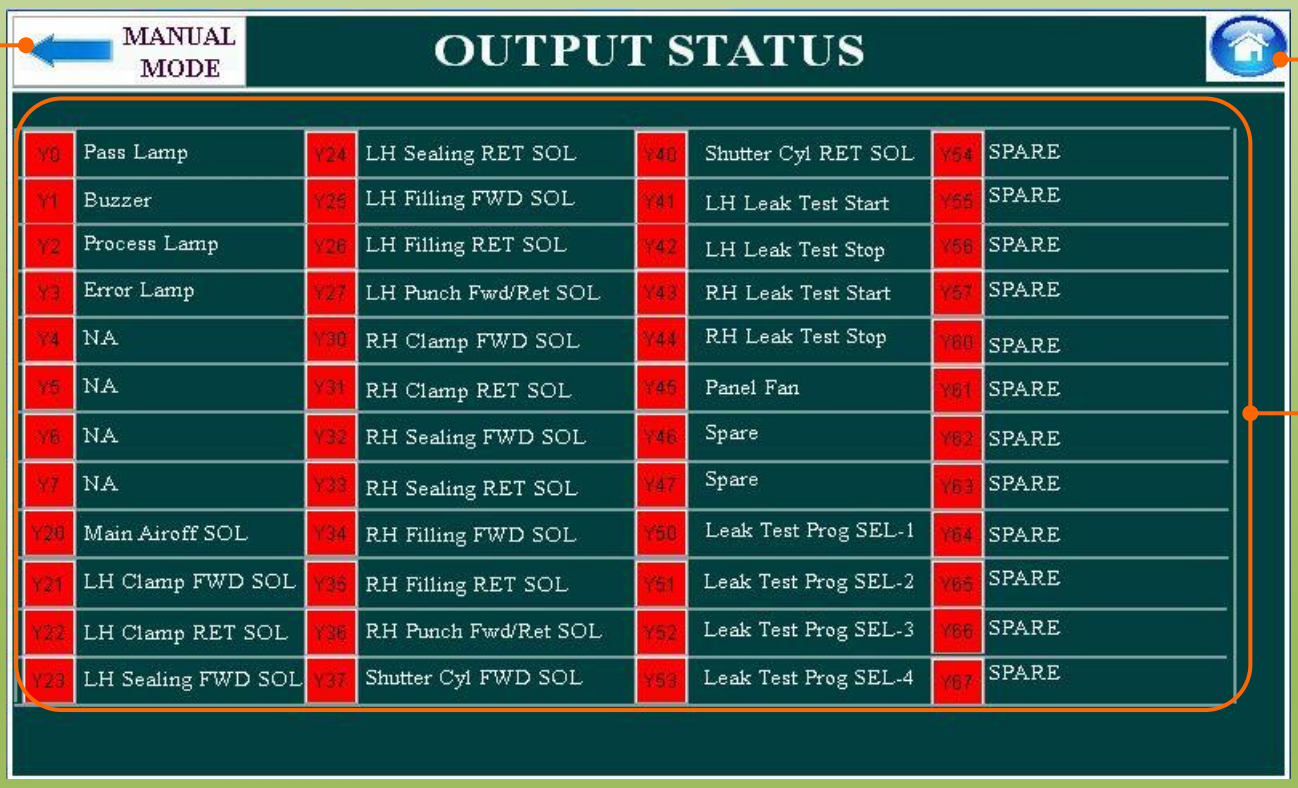
1. Minimize or maximize screen navigation option.
6. Home screen navigation button.
7. Programs list.
8. Program number display.
   - LH count reset navigation button.
   - RH count reset navigation button.

INPUT STATUS



1. Home screen navigation button.
2. Previous page navigation option.
3. Plc input status.
4. Indo details: -
   - LH yfg drl derivative navigation option.
   - RH yfg drl derivative navigation option.

OUTPUT STATUS



1. Home screen navigation button.
2. Previous page navigation option.
3. Plc output status.

MACHINE ERROR & REMENDIES:

SR.NO	ERROR DESCRIPTION		CAUSES OF ERROR		REMEDY
1	HOME ERROR!!!	**1**	There may be any pneumatic cylinder not in home position.	1	Set all retract reed sensor in correct position.
		2	The sensor position might have shifted.	2	Position the sensor properly.
		3	There may be loose connection in the sensor connector.	2	Tighten the sensor connector and press reset push button to clear the error.
2	EMERGENCY PB PRESSED	1	Emergency push button might have pressed.	1	Release emergency push button and press Reset push button to clear the error.
		2	There may be loose connection in the push button contact.	2	Tighten the electrical connection and press Reset push button to clear the error.
3	LOW AIR PRESSURE	1	Check the pneumatic supply.	1	Adjust the pressure to optimum level.
		2	There may be loose connection in the pressure indicator.	2	Tighten the connector and press Reset push button to clear the error.
4	FIXTURE ERROR	1	Check loaded fixture and set fixture in hmi are same or different.	1	Go in configuration screen then click on fixture number button to set actual fixture number.
		2	Check input indication on plc there may be loose connection in the electrical connector.	2	Tighten the electrical connection and press Reset push button to clear the error.
5	CLAMP CYL FWD ERROR	1	The sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
		3	The cylinder might not have reached its position fully.	3	Adjust the pressure to optimum level.
6	CLAMP CYL RET ERROR	1	The sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
		3	The cylinder might not have reached its position fully.	3	Adjust the pressure to optimum level.
7	PART SENSOR 1 ABSENT	1	The sensor position might have shifted.	1	Position the sensor properly.

		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
8	**PART SENSOR 2 ABSENT**	1	The sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
9	**PART SENSOR 3 ABSENT**	1	The sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
10	**PART SENSOR 4 ABSENT**	1	The sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
11	**PART SENSOR 5 ABSENT**	1	The sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
12	**SEALING CYL FWD ERROR**	1	The sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
13	**SEALING CYL RET ERROR**	1	The sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
14	**FILLING CYL FWD ERROR**	1	The sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
15	**FILLING CYL RET ERROR**	1	The sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
16	**LEAK TEST ERROR**	1	may be air leak from part	1	check the part ,do water deepn test
		2	may be air leak from seal and fill port	2	Check seal & fill port ,also check pneumatic tube
17	**DOT PUNCH CYL FWD ERROR**	1	The sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection	2	Tighten the sensor connector and press

			in the sensor connector		reset push button to clear the error
		3	The cylinder might not have reached its position fully.	3	Adjust the pressure to optimum level.
18	**DOT PUNCH CYL RET ERROR**	1	The sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
		3	The cylinder might not have reached its position fully.	3	Adjust the pressure to optimum level.
19	**BARCODE PRINTING ERROR**	1	Printing command not receive to printer	1	Check printer is on and in healthy condition
		2	There may be loose connection in the printer and plc communication cable	2	check communication cable
20	**CAMERA IMAGE-1 CHECK ERROR**	1	The part position might have shifted.	1	Position the part properly.
		2	Dirt may have accumulated on the part.	2	Clean part with cotton and check again.
21	**SHUTTER CYL FWD ERROR**	1	The sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
		3	The cylinder might not have reached its position fully.	3	Adjust the pressure to optimum level.
22	**SHUTTER CYL RET ERROR**	1	The sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
		3	The cylinder might not have reached its position fully.	3	Adjust the pressure to optimum level.
23	**NEW PART NOT PLACED**	1	After cycle complete the previous part are present still there.	1	Remove old part and place new part properly.
24	**CAMERA IMAGE-2 CHECK ERROR**	1	The part position might have shifted.	1	Position the part properly.
		2	Dirt may have accumulated on the part screen or part.	2	Clean part and check again
25	**LIGHT CURTAIN DISTURB**	1	There may be object or hand in between safety light curtain.	1	Remove the object or hand from the safety light curtain and press reset PB for reset error.

DO'S & DONT'S:

Every effort has been taken to ensure user safety while designing this machine. However, you are requested to adhere to following instructions to avoid any mishap.

- The machine should not be used for any other purpose other than that specified in the intended use.
- Do not operate the machine without understanding the installation, operation and maintenance manual thoroughly.
- Only trained, skilled, qualified and authorized personnel should install, operate and maintain the machine.
- Do not commission the machine without the aid of the qualified instructor.
- Do not reach into moving machinery.
- Never perform complex maintenance procedure single handed. Especially during start-ups as it is necessary to have another person manning the emergency stop in case of any problem.
- No tools other than that are prescribed should be used for machine maintenance.
- Do not store any loose tools in the machine compartments or control panel.
- If the machine is down for maintenance, place a sign over the control panel to inform others about that.
- Switch off the machine while connecting/disconnecting indo electrical connector.

CONTACT DETAILS

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